

# Solution Manual

## Chapter 1

### Introduction

#### Exercises

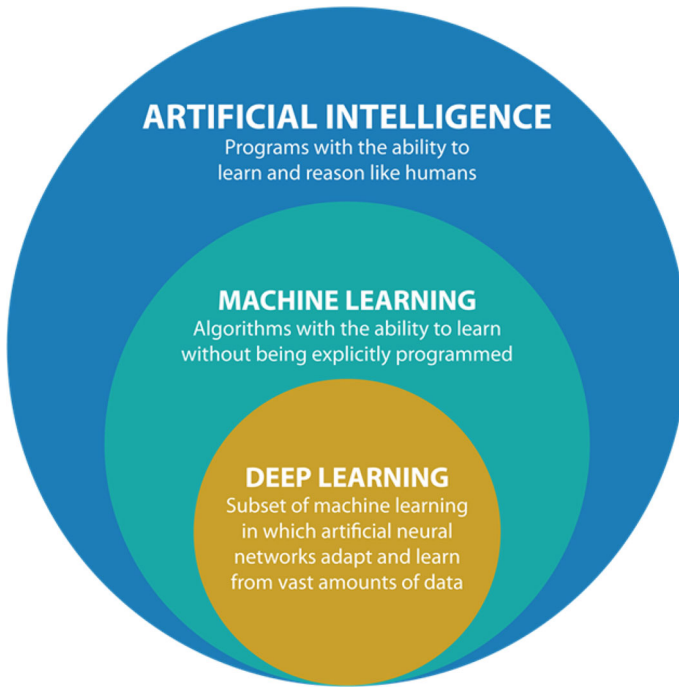
1.1 What is the relationship among AI (Artificial Intelligence), ML (Machine Learning), and DL (Deep Learning)?

1.1 Ans: The relationship among AI (Artificial Intelligence), ML (Machine Learning), and DL (Deep Learning) can be represented in Fig. A.1. Essentially, AI is an umbrella term for all forms of automation including learning and reasoning. ML is a subset of AI focusing on learning. DL is a specific tool of learning based on the artificial neural networks.

1.2 A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E. Suppose we feed a learning algorithm a lot of historical weather data, and have it learn to predict weather. In this setting what is T?

- (a) The historical weather data
- (b) The probability of it correctly predicting a future data's weather
- (c) The weather prediction task
- (d) None of these

1.2 Ans: (c)



**Fig. A.1** Relationship among artificial intelligence, machine learning and deep learning

1.3 What is data preprocessing and what are the typical processing tasks?

1.3 Ans: Data preprocessing is the process of cleaning and preparing data for learning and analysis. This can involve tasks such as removing invalid data, data imputation, and data normalization to prevent anomalies from dominating the learning and analysis.

1.4 In standardization, the data features will be re-scaled with

- (a) Mean 1 and Variance 1
- (b) Mean 0 and Variance 1
- (c) Minimum 0 and maximum 1
- (d) Minimum  $-1$  and maximum  $+1$

1.4 Ans: (b)

1.5 Which of the following is/are example(s) of feature extraction?

- (a) Measure the length, the width, the texture, and the number of fins of fishes for classification using computer vision
- (b) Imputation of missing data
- (c) Principal component analysis
- (d) Construct histogram of data values for each sample

1.5 Ans: (a,c,d) Reason:

- (a) The length, the width, the texture, and the number of fins of fishes can be considered data extracted from fishes.
- (c) The largest principal component subsets can be used as extracted features.
- (d) The histogram provides a statistical view of data from the binning perspective.

1.6 Suppose you are working on stock market prediction. Typically tens of millions of shares of a company's stock are traded each day. You would like to predict the number of shares that will be traded tomorrow for the company.

- (i) Would you treat this as a classification or a regression problem?
  - (a) Regression
  - (b) Classification
  - (c) Clustering
  - (d) None of these
- (ii) If the data you have collected involved millions of attributes, what would you do?

1.6 Ans: (i)(a), (ii)(Extract relevant features or perform a dimension reduction.)

1.7 Suppose you are working on weather prediction and use a learning algorithm to predict tomorrow's temperature (in degrees Centigrade/Fahrenheit).

- (i) Would you treat this as a classification or a regression problem?
  - (a) Regression
  - (b) Classification
  - (c) Clustering
  - (d) None of these.
- (ii) What kind of data should you gather?

1.7 Ans: (i) (a)

- (ii) Weather forecasts are made by collecting quantitative data (e.g., changes in barometric pressure, current weather conditions, and sky condition or cloud cover) about the current state of the atmosphere at a given place and using meteorology to project how the atmosphere will change.

1.8 What is/are the differences between supervised and unsupervised learning?

1.8 Ans: The algorithms of supervised learning use labeled data to get trained. The algorithms of unsupervised learning use unlabeled data for training purposes. In unsupervised learning, the models identify hidden data trends and do not take any feedback.

1.9 In a report, a research team claimed that their method is the best based solely on the low training error evaluation. The claim in the report is trustworthy. True or false?

1.9 Ans: The evaluation should use the unseen test error instead.

1.10 A research team analyzed information from several earlier studies involving more than 188 million of adults. They observed the link between sitting and risk of mortality over 20–30 years. In the studies, about 28% of deaths could be attributed to sitting, 21% to television viewing, the researchers said. The researchers concluded the analysis with the following statement: “Sit less than 5 years, add 2 years to your life”. This report is trustworthy as it involved a huge database. True or false?

1.10 Ans: This can be considered a causality creep since there is no proof for the causation.

## Chapter 2

### Data Preprocessing

#### Exercises

2.1 There is no difference between information and data. True or False?

2.1 Ans: False: Information is data that is processed in such a way that it becomes meaningful to the person who receives it, so information is not only a collection of raw facts.

2.2 For categorical data such as the type of fruits, the one-hot encoding is preferred over an arbitrary number assignment such as one for orange, two for apple, three for mango, etc. True or False?

2.2 Ans: Reason: Yes, this is because the higher values in arbitrary number assignment could have a higher influence for large numbers than that for small numbers. One-hot encoding does not have this problem.

2.3 Which of the following represents the characteristics of ordinal data?

- (a) Ordinal data has a median
- (b) Value of interval is unknown
- (c) Measures non-numeric traits such as satisfaction, happiness, etc.
- (d) Establish a relative rank
- (e) None of these

2.3 Ans: (a, b, c, d) and reasons:

- (a) Ordinal data has a median: Median is the value in the middle but not the middle value of a scale and can be calculated with data which has an innate order.
- (b) Value of interval is unknown: The ranking order does not have any scale or values in-between.

- (c) Measures non-numeric traits such as satisfaction, happiness, etc.: Measures quantitative levels of intensity qualitatively.
- (d) Establish a relative rank: Provides an order of intensity, strength, etc.

2.4 Determine whether the following items belong to interval or ratio data.

- (a) The number of hours one spent studying each day
- (b) The score of an intelligence test
- (c) The temperatures in cities throughout the country that are listed in most newspapers
- (d) The birth weights of babies who were born at Samsung Hospital last week
- (e) Ages of students in a machine learning class
- (f) Number of pages in your statistic book
- (g) The annual salaries for all teachers in a town
- (h) The cost of a pair of shoes

- 2.4 Ans: (a) The number of hours one spent studying each day (ratio scale)  
(b) The score of an intelligence test (ratio scale)  
(c) The temperatures in Fahrenheit (interval scale)  
(d) The birth weights of babies who were born at Samsung Hospital last week (ratio scale)  
(e) Ages of students in a machine learning class (ratio scale)  
(f) Number of pages in your statistic book (ratio scale)  
(g) The annual salaries for all teachers in a town (ratio scale)  
(h) The cost of a pair of shoes (ratio scale)

2.5 Which of the following is incorrect regarding the properties of distance metrics?

- (a) The distance from  $\mathbf{x}$  to  $\mathbf{y}$  is the same as the distance from  $\mathbf{y}$  to  $\mathbf{x}$
- (b) The metric must satisfy the triangular inequality
- (c) The distance between two identical vectors,  $\mathbf{x}$  and  $\mathbf{y}$ , is a non-zero value
- (d) The distance must never be negative
- (e) None of these

- 2.5 Ans: (c) The distance between two identical vectors,  $\mathbf{x}$  and  $\mathbf{y}$ , is a non-zero value. Reason: The distance between two identical vectors,  $\mathbf{x}$  and  $\mathbf{y}$ , is zero.

2.6 Which of the following distance measures calculates the distance between two binary vectors?

- (a) Euclidean distance
- (b) Manhattan distance
- (c) Minkowski distance
- (d) Hamming distance
- (e) None of these

2.6 Ans: (d) Hamming distance. Reason: The Hamming distance between two strings of equal length is the number of positions at which the corresponding symbols are different. In other words, it measures the minimum number of errors that could have transformed one string into the other.

2.7 What is the preferred way to handle missing or corrupted data in a small data set?

- (a) Drop the entire data sample that contains the corrupted features
- (b) Replace missing values with mean/median/mode
- (c) Replace missing features with zeros
- (d) Do not change the corrupted features
- (e) None of these

2.7 Ans: (b) Reason: Dropping the data sample would reduce the small data size further. Replace missing features with zeros or other arbitrary values would introduce bias to the data sample.

2.8 When the bounds of data are not known, the z-score standardization is preferred over min-max normalization. True or False?

2.8 Ans: True. Reason: This is because min-max normalization needs the boundary values for computing the standardized values of data.

2.9 You are given a set of data for supervised learning. A sample block of data looks like this:

|         |         |     |         |           |     |
|---------|---------|-----|---------|-----------|-----|
| 1.0234, | 0.3202, | 25, | 0.0081, | 60033.81, | 3   |
| 1.1356, | 0.3308, | 21, | 0.0067, | 69283.18, | -1  |
| 0.9988, | 0.2526, | 23, | 0.0093, | 66034.33, | 2   |
| 1.1858, | 0.3001, | 22, | 0.0077, | 64037.35, | 1   |
| 1.1533, | 0.3853, | 21, | 0.0066, | 62033.58, | -13 |
| 1.0755, | 0.3102, | 21, | 0.0098, | 60183.65, | 1   |
| 1.0045, | 0.2901, | 22, | 0.0065, | 61093.98, | -2  |
| 1.1031, | 0.3912, | 21, | 0.0088, | 69033.23, | -1  |

Each row corresponds to a sample data measurement with 5 input features and 1 response. What kind of undesired effect can you anticipate if this set of raw data is used for learning?

- (a) Those features with nominal values may create numerical instability
- (b) Those features with very large values may overshadow those with very small values
- (c) Those features with integer values may add bias to the system
- (d) The sample with output response value -13 should be imputed.
- (e) None of these

2.9 Ans: (b) Those features with very large values may overshadow those with very small values. We can either use min-max or z-score normalization to resolve the problem. (The output value of -13 does not seem abnormal for a regression problem)

- 2.10 Suppose you are given the following data regarding the monthly income of participants of a yoga learning group.

| Participants | a    | b    | c    | d    | e    | f    | g    | h      | i    |
|--------------|------|------|------|------|------|------|------|--------|------|
| Income \$    | 5000 | 3000 | 4000 | 4500 | 6000 | 3500 | 5000 | 800000 | 5500 |

Among mean, mode, and median, which is the most appropriate statistic to describe the data? Provide the value of this statistic based on the given table.

- 2.10 Ans: The median has better representation for this set of data.

Median: 3000, 3500, 4000, 4500, **5000**, 5000, 5500, 6000, 800000

## Chapter 3

### Linear Parametric Models

#### Exercises

3.1 Given  $\mathbf{X}\mathbf{w} = \mathbf{y}$  where  $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 5 & 6 \end{bmatrix}$ ,  $\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ .

- (a) What kind of system is this? (even-, over- or under-determined?)  
 (b) Is  $\mathbf{X}$  invertible? Why?  
 (c) Solve for  $\mathbf{w}$  if it is solvable.

- 3.1 Ans: (a) This is an even-determined system because it has the same the number of equations and the number of unknowns.

- (b)  $\mathbf{X}$  is invertible since  $\det(\mathbf{X}) = 1 \times 6 - 1 \times 5 = 1 \neq 0$ .

$$\mathbf{X}^{-1} = \begin{bmatrix} 6 & -1 \\ -5 & 1 \end{bmatrix}.$$

(c)  $\hat{\mathbf{w}} = \mathbf{X}^{-1}\mathbf{y} = \begin{bmatrix} 6 & -1 \\ -5 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$

- 3.2 The linear system given by  $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$  is over-determined and thus has no exact solution. True or False?

- 3.2 Ans: False. Reason: since the  $\text{rank}(\mathbf{X}) = \text{rank}([\mathbf{X}|\mathbf{y}]) = 2$ , it has an exact solution.

3.3 Consider the following system:

$$\begin{aligned}w_1 + w_2 &= 0 \\w_1 - w_2 + w_3 - w_4 &= 1 \\w_3 + w_4 &= 2\end{aligned}$$

What kind of system is this? Can the system be solved? Select one or more from below:

- (a) The system is even-determined and can be solved with an exact solution.
- (b) The system is over-determined and can be solved with an approximated solution.
- (c) The system is over-determined and it does not have an exact solution.
- (d) The system is under-determined and can be solved with an exact least norm solution.
- (e) The system is under-determined and it has infinite number of solutions.

3.3 Ans: (d, e) An under-determined system with full row rank has infinite number of solutions. The least norm solution is unique and is given by

$$\hat{\mathbf{w}} = \mathbf{X}^T (\mathbf{X}\mathbf{X}^T)^{-1} \mathbf{y} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 1 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0.25 \\ -0.25 \\ 1.25 \\ 0.75 \end{bmatrix}.$$

3.4 Identify those polynomial models shown below which include at least a second-order term:

- (a)  $1 + x_1 + x_2 + x_1x_2 - x_1^2 - x_2^2$
- (b)  $1 + x_1 + x_2 + x_3 + x_4 + x_5$
- (c)  $1 + x_1 + x_2 + x_1x_2$
- (d)  $1 + x_1 + x_2 + x_1(x_1 - x_2)$
- (e) None of these

3.4 Ans: (a, c, d) Reason: The degree of a polynomial is the highest degree of its monomials. You can find the degree by adding up the exponents of the variables that appear in it. For example, the polynomial  $x^2y^3 + xy + 4x^3y^4$  has 3 terms. The first term has a degree of  $2 + 3 = 5$ ; The second term has a degree of  $1 + 1 = 2$ ; The third term has a degree of  $4 + 3 = 7$ ; Therefore the polynomial has a degree of 7 which is the highest degree of any term.

3.5 Given the input feature  $x$  and the corresponding response  $y$  as follows.

|     |     |    |    |    |   |   |
|-----|-----|----|----|----|---|---|
| $x$ | -10 | -7 | -3 | -1 | 4 | 8 |
| $y$ | 10  | 11 | 8  | 6  | 3 | 4 |

- (a) Perform a third-order polynomial regression and plot the result of line fitting.  
 (b) Given a test point  $x = -5$  predict  $y$  using the learned polynomial model.

3.5 Ans: (a) Polynomial of third-order for 1D feature:  $g(\mathbf{x}) = w_0 + w_1x + w_2x^2 + w_3x^3$ . Pack all the data points to form the polynomial matrix

$$\mathbf{P} = \begin{bmatrix} 1 & -10 & (-10)^2 & (-10)^3 \\ 1 & -7 & (-7)^2 & (-7)^3 \\ 1 & -3 & (-3)^2 & (-3)^3 \\ 1 & -1 & (-1)^2 & (-1)^3 \\ 1 & 4 & (4)^2 & (4)^3 \\ 1 & 8 & (8)^2 & (8)^3 \end{bmatrix} \quad \text{with } \mathbf{y} = \begin{bmatrix} 10 \\ 11 \\ 8 \\ 6 \\ 3 \\ 4 \end{bmatrix}.$$

This is an over-determined system with solution given by

$$\hat{\mathbf{w}} = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \mathbf{y} = \begin{bmatrix} 5.3280 \\ -0.8977 \\ 0.0330 \\ 0.0075 \end{bmatrix}$$

(b) Prediction:  $\hat{y} = \mathbf{p}_i^T \hat{\mathbf{w}} = [1, -5, (-5)^2, (-5)^3]^T \hat{\mathbf{w}} = 9.7081$  (Fig. A.2).

3.6 This question is also about polynomial models. Which of the following models has discriminative feature for the data given by

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} +1 \\ +1 \end{bmatrix}, \begin{bmatrix} -1 \\ +1 \end{bmatrix}, \begin{bmatrix} +1 \\ -1 \end{bmatrix}$$

with corresponding target values  $y \in \{+1, +1, -1, -1\}$ ?

- (a)  $g(\mathbf{x}) = 1 + w_1x_1 + w_2x_2$   
 (b)  $g(\mathbf{x}) = 1 + w_1x_1 + w_2x_2 + w_3x_1x_2$   
 (c)  $g(\mathbf{x}) = 1 + w_1x_1 + w_2x_2 + w_3x_1^2$

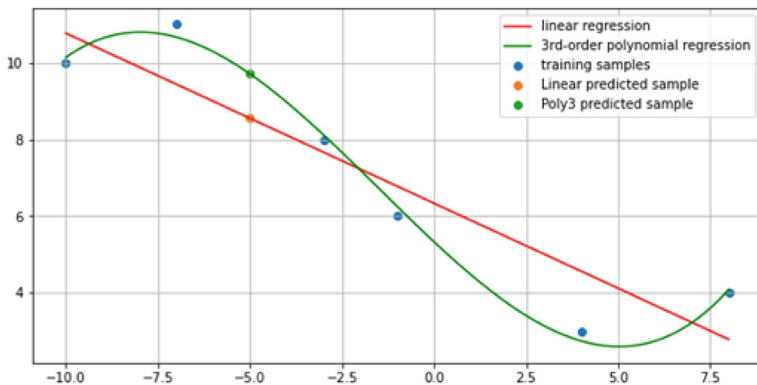


Fig. A.2 The fitting curve

- (d)  $g(\mathbf{x}) = 1 + w_1x_1 + w_2x_2 + w_3x_2^2$   
 (d)  $g(\mathbf{x}) = 1 + w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_2^2$   
 (e) None of these

3.6 Ans: (b)

Reason: By substituting the  $\mathbf{x}$  values into the monomial terms of the given functions, we have

- $x_1x_2 : \{+1, +1, -1, -1\}$  (This term is discriminative since it has good alignment with the target values)
- $x_1^2 : \{+1, +1, +1, +1\}$  (This term is not discriminative)
- $x_2^2 : \{+1, +1, +1, +1\}$  (This term is not discriminative)

Hence the only discriminative function is (b) which contains discriminative feature  $x_1x_2$ .

3.7 This question is also about multivariate polynomials.

- (a) Write down the expression for a third-order polynomial model having a three-dimensional input.  
 (b) Write down the  $\mathbf{P}$  matrix for this polynomial given

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix}.$$

- (c) Given  $\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ , can a unique solution be obtained for this under-determined system of  $\mathbf{P}$  matrix?

3.7 Ans: (a) A polynomial model of third order can be written as  $g(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_1x_2 + w_5x_2x_3 + w_6x_1x_3 + w_7x_1^2 + w_8x_2^2 + w_9x_3^2 + w_{10}x_2x_1^2 + w_{11}x_3x_1^2 + w_{12}x_1x_2^2 + w_{13}x_3x_2^2 + w_{14}x_1x_3^2 + w_{15}x_2x_3^2 + w_{16}x_1x_2x_3 + w_{17}x_1^3 + w_{18}x_2^3 + w_{19}x_3^3$ .

- (b) The polynomial expansion with data is written in matrix form as

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & -1 & 1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & 1 & -1 & 1 \end{bmatrix}.$$

- (c) Yes, a unique solution can be obtained (using Python) for the under-determined system as follows:  
 $\hat{\mathbf{w}} = \mathbf{P}^T(\mathbf{P}\mathbf{P}^T)^{-1}\mathbf{y} = [0, 0, -0.1, 0, -0.1, -0.1, 0, 0, 0.1, 0, -0.1, 0, 0.1, 0.1, 0, -0.1, -0.1, 0, -0.1, 0].$

3.8 This question is about the reduced multivariate polynomials.

- (a) Write down the expression for a third-order RM model having a three-dimensional input.  
 (b) Write down the  $\mathbf{P}$  matrix for this polynomial given

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix}.$$

- (c) Given  $\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ , can a unique solution be obtained for this under-determined system of  $\mathbf{P}$  matrix?

3.8 Ans: (a) A RM model of third order can be written as  $g(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_1^2 + w_5x_2^2 + w_6x_3^2 + w_7x_1^3 + w_8x_2^3 + w_9x_3^3 + w_{10}x_1(x_1 + x_2 + x_3) + w_{11}x_2(x_1 + x_2 + x_3) + w_{12}x_3(x_1 + x_2 + x_3) + w_{13}x_1(x_1 + x_2 + x_3)^2 + w_{14}x_2(x_1 + x_2 + x_3)^2 + w_{15}x_3(x_1 + x_2 + x_3)^2$ .

(b) The reduced polynomial expansion with data is written in matrix form as

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 2 & 0 & 2 & 4 & 0 & 4 \\ 1 & 1 & -1 & 1 & 1 & 1 & 1 & 1 & -1 & 1 & 1 & -1 & 1 & 1 & -1 & 1 \end{bmatrix}.$$

(c) Yes, a unique solution can be obtained for the under-determined system as follows:

$$\hat{\mathbf{w}} = \mathbf{P}^T (\mathbf{P}\mathbf{P}^T)^{-1} \mathbf{y} = [0.0716, 0.0716, -0.1202, 0.0716, 0.0716, 0.1202, 0.0716, 0.0716, -0.1202, 0.0716, 0.0230, -0.1202, 0.0230, -0.0742, -0.1202, -0.0742].$$

3.9 The two-layer perceptron network with linear output layer can be written in two ways, namely

$$g(\mathbf{W}_o, \mathbf{W}_h, \mathbf{x}) = \mathbf{W}_o L(\mathbf{W}_h \mathbf{x}), \quad \mathbf{W}_h \in \mathbb{R}^{h \times d}, \quad \mathbf{W}_o \in \mathbb{R}^{o \times h},$$

and

$$g(\mathbf{W}_o, \mathbf{W}_h, \mathbf{x}) = L(\mathbf{x}^T \mathbf{W}_h) \mathbf{W}_o, \quad \mathbf{W}_h \in \mathbb{R}^{d \times h}, \quad \mathbf{W}_o \in \mathbb{R}^{h \times o},$$

where  $L(\cdot) = \max(0, \cdot)$  for  $\mathbf{x} \in \mathbb{R}^{d \times 1}$ . Let's use the second form for our computation and assume  $\mathbf{W}_h$  has been fixed at

$$\mathbf{W}_h = \begin{bmatrix} 2 & 2 & 3 \\ 3 & 0 & 2 \end{bmatrix}.$$

Suppose you are given several training samples  $\mathbf{x}$  with corresponding target  $\mathbf{y}$ :

$$\mathbf{x} : \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}, \quad \mathbf{y} : \{0, 0, 1, 1\}.$$

Can the output weights  $\mathbf{W}_o$  be solved deterministically?

3.9 Ans: Yes, the solution is over-determined which is given by

$$\hat{\mathbf{W}}_o = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \mathbf{y} = [0.25, 0, 0.5]^T$$

where  $\mathbf{P} = L(\mathbf{x}^T \mathbf{W}_h)$  and  $\mathbf{y} = [0, 0, 1, 1]^T$ .

1. Codes:

$$\mathbf{X} = [1 \ 0; \ 0 \ 1; \ 0 \ 0; \ 1 \ 1]$$

$$\mathbf{Y} = [0; \ 0; \ 1; \ 1]$$

$$\mathbf{W}_h = [2 \ 2 \ 3; \ 3 \ 0 \ 2]$$

$$\mathbf{P} = \text{ReLU}(\mathbf{X} * \mathbf{W}_h)$$

```
w = inv(P'*P)*P'*y

function p = ReLU(x)
    p = max(0,x);
end
```

3.10 Code a function to build a regressor matrix for the Reduced Multivariate Polynomials Reciprocal Sigmoid features within MATLAB/Octave.

3.10 Ans: Codes:

```
function P = RMRsigmoid(X,order)
% Build regressor matrix P (mxK):
% order = desired order of approximation,
% X = input matrix (mx1), K = number of parameters to be est.
% m = number of data samples, l = input dimension.
[m,l] = size(X); MM1=[]; MM3=[]; Msum=sum(X,2);
X = 1 + exp(-X); % Reciprocal Sigmoid features
for i=1:order
    for k=1:l
        M1(:,k)=X(:,k).^i;
        if (i>1)
            M3(:,k)=X(:,k).*Msum.^(i-1);
        end
    end
    MM1=[MM1,M1];
    if (i>1)
        MM3=[MM3,M3];
    end
end
P = [ones(m,1),MM1,MM3];
return;
```

## Chapter 4

### Learning Score Functions

#### Exercises

4.1 What is the difference between a regression loss and a classification loss?

4.1 Ans: The regression loss involves prediction of real valued, continuous target values. The classification loss involves prediction of discrete target

category or class and label. While regression loss predicts values, classification loss predicts target classes or labels.

4.2 What is the difference between a loss function and an error function?

4.2 Ans: An error function measures the deviation of an observable value from a prediction. A loss function works on an error to quantify its negative consequences of error. For example, the regression loss measures the squared error distance between the predicted value and the target value. The classification loss measures the number of samples being classified incorrectly based on the error observed or deviation of an observable value from a prediction.

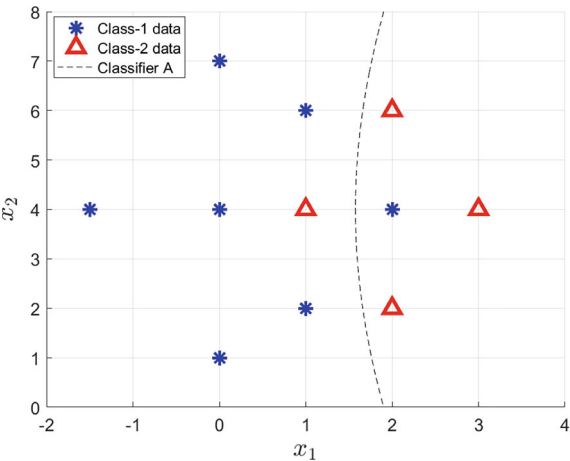
4.3 What is the difference between a loss function and a cost function?

4.3 Ans: A loss function quantifies the negative consequences of error for a single training sample, and the cost function aggregates these losses from the training samples to form a measurement objective. The cost function is also known as the objective function, which is a function specified for either minimization or maximization. The objective function is a mathematical optimization term that embodies the set of functions being optimized and the constraints involved in achieving the optimization.

4.4 What are the common loss functions for regression and that for classification?

4.4 Ans: Common losses for regression include the Mean-Squared Error (MSE) and Mean Absolute Error (MAE). Common losses for classification include the Margin Loss, Cross-entropy loss, and 0–1 loss.

4.5 The plot below shows the decision boundary of Classifier A. Write down the confusion matrix for Classifier A below.



4.5 Ans: Confusion matrix

|                   | Class-1 | Class-2 |
|-------------------|---------|---------|
| Predicted Class-1 | 6       | 1       |
| Predicted Class-2 | 1       | 3       |

4.6 Consider the experimental observations of the Training error rate  $Tr$  and the Validation error rate  $Va$  for a machine learning algorithm for each setting of the parameter  $\theta$ . Choose a parameter ( $P$ ) based on  $Tr$  and  $Va$ .

| $\theta$ | Tr   | Va   |
|----------|------|------|
| 0.1      | 0.08 | 0.23 |
| 0.2      | 0.33 | 0.38 |
| 0.3      | 0.21 | 0.18 |
| 0.4      | 0.13 | 0.26 |
| 0.5      | 0.16 | 0.18 |

Which value of parameter  $\theta$  will you choose based on the above observation?

4.6 Ans:  $\theta = 0.5$

Reason: There are two cases with lowest  $Va$  value: at  $\theta = 0.3$  and at  $\theta = 0.5$ . The former case shows more underfit in training than that of the latter case.

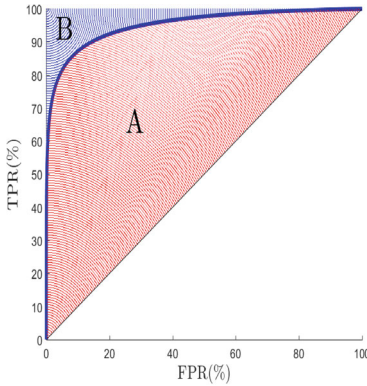
4.7 Consider the binary classification problem, which you are dealing with, has highly imbalanced classes. The majority class has 99 hundred samples and the minority class has 1 hundred samples. Which of the following metric(s) would you choose for assessing the classification performance? (Select all relevant metric(s))

- (a) Accuracy
- (b) Cost sensitive accuracy
- (c) Precision and recall
- (d) Type-I and type-II errors
- (d) None of these

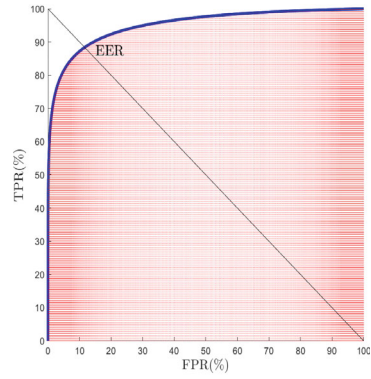
4.7 Ans: The answers are (b, c, d).

Reason: A cost sensitive accuracy is needed here for imbalance classes. The choices (b, c, d) show distinct results for each of the two classes (such as minority/majority, type-I/type-II or positive/negative classes).

4.8 According to the plots of Figs. 4.8b and 4.10 (reproduced below), what is the relationship between the Gini Coefficient and the Area Under the ROC (AUC)?



(a) Gini



(b) ROC

4.8 Ans: From plot (a), the Gini-coefficient =  $A/(A+B)$ . Since the area  $(A+B)$  is  $1/2$ ,  $Gini = A/(1/2) = 2A$ .

From the ROC area in plot (b),  $AUC = A + 1/2 \Rightarrow A = AUC - 0.5$ .

Substitute  $A$  into the Gini above:  $Gini = 2(AUC - 0.5)$ .

4.9 Given the following confusion matrix for a binary classifier:

|                     | Class(+1) | Class(-1) |
|---------------------|-----------|-----------|
| Predicted Class(+1) | 990       | 20        |
| Predicted Class(-1) | 10        | 380       |

Let  $n^+$  and  $n^-$  denote the numbers of Class(+1) and Class(-1) samples respectively. Calculate the

- True Positive rate;
- True Negative rate;
- Accuracy at  $s = n^-/n^+$  (this is the most widely adopted accuracy);
- Accuracy at  $s = 1$ ;
- Precision at  $s = n^-/n^+$ .

4.9 Ans: (a) True Positive rate:  $TPR = TP/n^+ = 990/(990 + 10) = 0.99$ .

(b) True Negative rate:  $TNR = TN/n^- = 380/(380 + 20) = 0.95$ .

(c) Accuracy at  $s = n^-/n^+$ :  $(TP + TN)/(n^+ + n^-) = (990 + 380)/(1000 + 400) = 0.9786$ .

(d) Accuracy at  $s = 1$ :  $(TPR + TNR)/2 = (0.99 + 0.95)/2 = 0.97$ .

(e) Precision at  $s = n^-/n^+$ :  $TP/(TP + FP) = 990/(990 + 20) = 0.9802$ .

4.10 The Misclassification Error (MCE) is the count of samples classified incorrectly. Suppose the binary targets are labeled as  $\{-1, +1\}$  and  $g(\mathbf{x})$  denotes the predictor output. Express the MCE in terms of the margin  $y_i g(\mathbf{x}_i)$ .

4.10 Ans: For each sample, the margin  $y_i g(\mathbf{x}_i)$  is positive if the sample is classified correctly and negative vice versa. The MCE is the total number of samples classified incorrectly. Using  $\text{sgn}(y_i g(\mathbf{x}_i))$  where  $\text{sgn}(\cdot) \in \{-1, +1\}$ , we only need to count those samples which give negative margins. To obtain a counting with positive numbers, we need to invert the sign of the counting as follows:

$$\text{MCE} = - \sum_{i=1}^n \min\{0, \text{sgn}(y_i g(\mathbf{x}_i))\}.$$

## Chapter 5

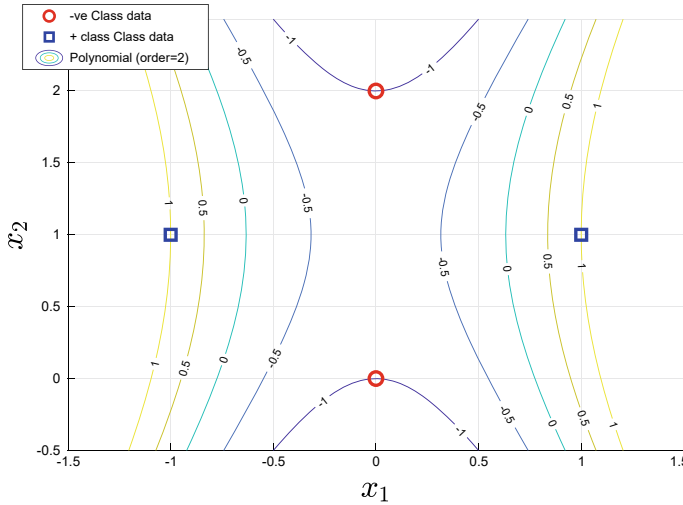
### Analytic Learning

#### Exercises

5.1 The polynomial function given by  $g(\mathbf{x}, \mathbf{w}) = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_1 x_2 + w_4 x_1^2 + w_5 x_2^2$  is linear with respect to the input feature vector  $\mathbf{x} = [x_1, x_2]^T$ , and hence the learning of its parameters can be analytic. True or False?

5.1 Ans: False. Reason:  $g(\mathbf{x}, \mathbf{w})$  is NOT linear with respect to the input feature vector  $\mathbf{x} = [x_1, x_2]^T$ , but it is linear with respect to  $\mathbf{w}$ .

5.2 The contours of decision surface in the following plot have been produced by learning a second-order polynomial model based on four data points drawn from two categories. Which of the following statement(s) is/are true?



- (a) The system is an over-determined one
- (b) The system is an under-determined one
- (c) The system is an even-determined one
- (d) The learned solution is exact with no approximation
- (e) None of these

5.2 Ans: (b) and (d).

Reason: The second-order polynomial expansion on two features ( $x_1$  and  $x_2$  of the plot) give rise to model  $g(\mathbf{x}, \mathbf{w}) = w_0 + w_1x_1 + w_2x_2 + w_3x_1x_2 + w_4x_1^2 + w_5x_2^2$  which has 6 parameters. Using this polynomial to learn 4 data points in the plot constitutes an under-determined system. The learning is thus exact with no approximation.

5.3 Consider the optimization problem given by

$$\min_{\mathbf{w}} J(\mathbf{w}) = \min_{\mathbf{w}} \left\{ \sum_{i=1}^m (f(\mathbf{w}, \mathbf{x}_i) - y_i)^2 + \lambda \mathbf{w}^T \mathbf{w} \right\}$$

where  $f(\mathbf{w}, \mathbf{x}_i) = \mathbf{x}_i^T \mathbf{w}$  with each  $\mathbf{x}_i$  being non-zero independent entry,  $\mathbf{w} \in \mathbb{R}^{(d+1) \times 1}$  is the parameter, each  $y_i \in \mathbb{R}$  is the learning target. When the covariance  $\mathbf{X}^T \mathbf{X}$  is invertible, a least squares solution exists in the form of  $\hat{\mathbf{w}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ . Which of the following statement(s) is/are correct?

- (a) The first derivative of  $f(\mathbf{w}, \mathbf{x}_i)$  with respect to  $\mathbf{w}$  at  $\hat{\mathbf{w}}$  must be zero.
- (b) The first derivative of  $f(\mathbf{w}, \mathbf{x}_i)$  with respect to  $\mathbf{w}$  at  $\hat{\mathbf{w}}$  need not be zero.
- (c) The second derivative of  $f(\mathbf{w}, \mathbf{x}_i)$  with respect to  $\mathbf{w}$  at  $\hat{\mathbf{w}}$  must be zero.
- (d) The second derivative of  $f(\mathbf{w}, \mathbf{x}_i)$  with respect to  $\mathbf{w}$  at  $\hat{\mathbf{w}}$  must be positive definite.
- (e) Non of these.

5.3 Ans: (b) Reason:

- For setting the derivative to 0, it concerns the cost function  $J$  and not the learning model  $f(\mathbf{w}, \mathbf{x}_i)$ !
- Nevertheless, the first derivative of the learning model  $f(\mathbf{w}, \mathbf{x}_i) = \mathbf{x}_i^T \mathbf{w}$  wrt  $\mathbf{w}$  is  $\mathbf{x}_i$  and it is not necessarily 0.

5.4 Consider the following data samples with  $x_1$  and  $x_2$  being the data features, and  $y$  being the learning target output class label.

|       |    |    |    |    |
|-------|----|----|----|----|
| $x_1$ | 0  | 1  | 1  | 0  |
| $x_2$ | 0  | 1  | 0  | 1  |
| $y$   | -1 | -1 | +1 | +1 |

Without regularization, construct a polynomial model of second-order to learn these data and then predict the test output labels of  $(x_1, x_2) \in \{(0.1, 0.1), (0.9, 0.9), (0.9, 0.1), (0.1, 0.9)\}$ .

5.4 Ans: The second-order polynomial model can be written as  $g(x_1, x_2) = w_0 + w_1x_1 + w_2x_2 + w_3x_1x_2 + w_4x_1^2 + w_5x_2^2$  and the corresponding polynomial regressor matrix is

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}.$$

This is an under-determined system since there are more parameters than sample size. The solution for the parameters is given by

$$\begin{aligned} \hat{\mathbf{w}} &= \mathbf{P}^T (\mathbf{P}\mathbf{P}^T)^{-1} \mathbf{y} \\ &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} -1 \\ -1 \\ +1 \\ +1 \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ 1 \\ 1 \\ -4 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

The test label predictions are

$$\hat{g}_{class}(x_1, x_2) = \text{sgn}(\mathbf{P}_t \hat{\mathbf{w}}) = \text{sgn} \left( \begin{bmatrix} -0.82 \\ -0.82 \\ 0.46 \\ 0.46 \end{bmatrix} \right) = \begin{bmatrix} -1 \\ -1 \\ +1 \\ +1 \end{bmatrix}.$$

5.5 Consider the following data samples with  $x_1$  and  $x_2$  being the data features, and  $y$  being the learning target output value.

|       |      |      |      |     |     |     |     |     |     |     |     |     |
|-------|------|------|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| $x_1$ | 0    | 0    | 1    | 2   | 0   | 1   | 3   | 2   | 4   | 5   | 3   | 6   |
| $x_2$ | 0    | 1    | 0    | 1   | 2   | 2   | 1   | 3   | 5   | 3   | 6   | 8   |
| $y$   | -2.1 | -1.3 | -1.5 | 1.0 | 0.5 | 0.8 | 2.1 | 1.8 | 3.3 | 2.8 | 3.8 | 5.0 |

First construct a polynomial model of third-order without regularization to learn these data and then predict the test outputs of  $(x_1, x_2) \in \{(0.5, 0.3), (0.8, 0.8)\}$ . Next learn the same third-order polynomial model with regularization at  $\lambda = 0.0001$  for predicting the same test points.

5.5 Ans: The third-order polynomial model can be written as  $g(x_1, x_2) = w_0 + w_1x_1 + w_2x_2 + w_3x_1x_2 + w_4x_1^2 + w_5x_2^2 + w_6x_1^2x_2 + w_7x_2^2x_1 + w_8x_1^3 + w_9x_2^3$ . The corresponding  $\mathbf{P}$  matrix incorporating the training data can be written as

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 1 & 2 & 4 & 1 & 4 & 2 & 8 & 1 \\ 1 & 0 & 2 & 0 & 0 & 4 & 0 & 0 & 0 & 8 \\ 1 & 1 & 2 & 2 & 1 & 4 & 2 & 4 & 1 & 8 \\ 1 & 3 & 1 & 3 & 9 & 1 & 9 & 3 & 27 & 1 \\ 1 & 2 & 3 & 6 & 4 & 9 & 12 & 18 & 8 & 27 \\ 1 & 4 & 5 & 20 & 16 & 25 & 80 & 100 & 64 & 125 \\ 1 & 5 & 3 & 15 & 25 & 9 & 75 & 45 & 125 & 27 \\ 1 & 3 & 6 & 18 & 9 & 36 & 54 & 108 & 27 & 216 \\ 1 & 6 & 8 & 48 & 36 & 64 & 288 & 384 & 216 & 512 \end{bmatrix}.$$

Based on this setting, the solutions without and with regularization are respectively  $\hat{\mathbf{w}}_a = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \mathbf{y}$  and  $\hat{\mathbf{w}}_b = (\mathbf{P}^T \mathbf{P} + \lambda \mathbf{I})^{-1} \mathbf{P}^T \mathbf{y}$  where  $\mathbf{y} = [-2.1, -1.3, -1.5, 1.0, 0.5, 0.8, 2.1, 1.8, 3.3, 2.8, 3.8, 5.0]^T$ .

The learned parameters for non-regularized and regularized cases are

$$\hat{\mathbf{w}}_a = [-2.4096, 0.8002, 1.6881, -0.4563, 0.4047, -0.2161, 0.0959, -0.0433, -0.0913, 0.0371]^T$$

and

$$\hat{\mathbf{w}}_b = [-2.4093, 0.8001, 1.6879, -0.4563, 0.4047, -0.2160, 0.0958, -0.0433, -0.0912, 0.0371]^T$$

respectively.

The prediction the two test points are

$\hat{\mathbf{y}}_a = \mathbf{P}_t \hat{\mathbf{w}}_a = [-1.4949, -0.5910]^T$  and  $\hat{\mathbf{y}}_b = \mathbf{P}_t \hat{\mathbf{w}}_b = [-1.4948, -0.5910]^T$  for the non-regularized case and the regularized case respectively.

Depending on the sensitivity of inverse computation, the non-regularized case might encounter “matrix close to singular or badly scaled, results may be inaccurate.” The two learning results also show that the regularized case gives an estimation with a smaller norm value.

- 5.6 You are given a collection of 6 training data points of two features ( $\mathbf{x} = [x_1, x_2]^T$ ) and their class labels ( $y \in \{-1, +1\}$ ) as follows:

$$y = -1 : \mathbf{x} \in \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0.2 \end{bmatrix}, \begin{bmatrix} 0.2 \\ 0.2 \end{bmatrix}, \begin{bmatrix} 0.2 \\ 0 \end{bmatrix} \right\},$$

$$y = +1 : \mathbf{x} \in \left\{ \begin{bmatrix} 0.2 \\ 0.1 \end{bmatrix}, \begin{bmatrix} 0.3 \\ 0.1 \end{bmatrix} \right\}.$$

These data have been used for training a partial polynomial model of third-order given by equation (1) utilizing the criterion function given by equation (2) below.

$$g(\mathbf{w}, \mathbf{x}) = w_0 + \sum_{i=1}^2 \sum_{j=1}^2 w_{ij} x_i x_j + \sum_{i=1}^2 \sum_{j=1}^2 \sum_{k=1}^2 w_{ijk} x_i x_j x_k, \quad (1)$$

$$\min_{\mathbf{w}} J(\mathbf{w}) = \min_{\mathbf{w}} \sum_{i=1}^m (g(\mathbf{w}, \mathbf{x}) - y_i)^2. \quad (2)$$

How many of the above samples are incorrectly classified by the partial polynomial model incorporating a signum threshold?

- 5.6 Ans: The dual solution without regularization is  $\hat{\mathbf{w}} = \mathbf{P}^T (\mathbf{P}\mathbf{P}^T)^{-1} \mathbf{y}$  and the classified output is  $\hat{\mathbf{y}} = \text{sgn}(\mathbf{P}\hat{\mathbf{w}}) = [-1, -1, -1, -1, +1, +1]^T$ .

Utilizing Python for the computation:

```
import numpy as np
from numpy.linalg import inv
from sklearn.preprocessing import PolynomialFeatures

X = np.array([[0,0],[0,0.2],[0.2,0.2],[0.2,0],[0.2,0.1],[0.3,0.1]])
y = np.array([-1, -1, -1, -1, 1, 1])
poly = PolynomialFeatures(3)
Pfull = poly.fit_transform(X)
Ppart = Pfull[:,3:10]
P = np.column_stack((np.ones((len(Ppart), 1)), Ppart))
if (P.shape[1] > P.shape[0]): # dual
    w = P.T @ inv(P @ P.T) @ y
else: # primal
    w = (inv(P.T @ P) @ P.T) @ y
```

```

y_trained = P @ w
y_class = np.sign(y_trained)
print(y_class)

```

5.7 Generate the data points using the MATLAB codes below:

```

x = [0.1:.1:10];
rand('state',8);
y = [0.6+0.05*rand(1,length(x)/2), 0.3+0.05*rand
(1,length(x)/2)]';

```

Based on the knowledge that a kernel measures some kind of distance between two data points, construct the kernel matrix corresponding to  $K(i, j) = e^{\frac{1}{2\sigma^2} \|x_i - x_j\|_2^2}$  for two  $\sigma$  values given by 0.1 and 1. Learn the above data by kernel ridge regression with the two given  $\sigma$  values and use the two learning results to predict test samples given by

```

x_ts = [0.1:.01:10.05];

```

What do you observe regarding the difference between the two predictions?

5.7 Ans: Firstly the learning kernel matrix can be constructed by

```

% Form Training Kernel
sigma = 1;
for i = 1:length(x)
    for j = 1:length(x)
        K(i,j) = exp(-(x(i)-x(j))^2/(2*sigma^2));
    end
end

```

The value of sigma can be changed to 1 to form another kernel matrix.

Next, learn the data using the kernel ridge regression as follows.

```

% Training
lambda = 0.001;
alpha = inv(K + lambda*eye(size(K)))*y;

For the test prediction, form the test kernel matrix and then use it for prediction as follows.

% Form Test Kernel
x_ts = [0.1:.01:10.05]; % Test data
for i = 1:length(x_ts)
    for j = 1:length(x)
        Kts(i,j) = exp(-(x_ts(i)-x(j))^2/(2*sigma^2));
    end
end

% Test output
y_ts = Kts*alpha;

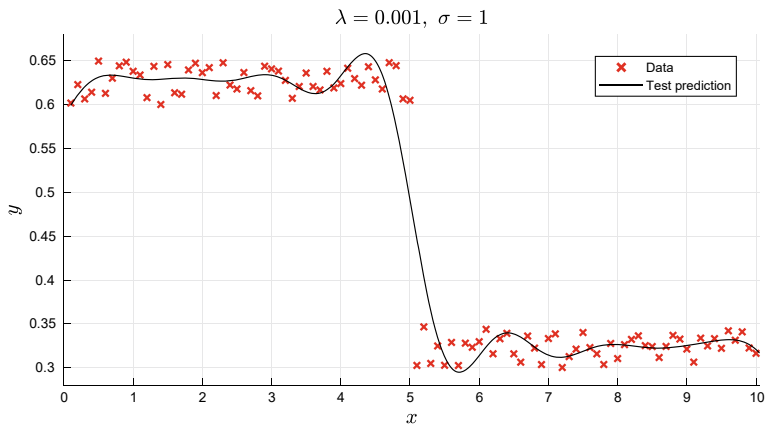
Plotting the results:
figure(1), hold on
h1=plot(x,y,'xr','MarkerSize',8,'LineWidth',2);
h2=plot(x_ts,y_ts,'-k','LineWidth',1);
xlabel('$x$', 'Interpreter', 'latex', 'FontSize', 18)

```

```

ylabel('$y$', 'Interpreter', 'latex', 'FontSize', 18)
% title('$\lambda=0.001, \sigma=1$', 'Interpreter', 'latex', 'FontSize', 18)
title('$\lambda=0.001, \sigma=0.1$', 'Interpreter', 'latex', 'FontSize', 18)
lgd=legend([h1, h2], 'Data', 'Test prediction');
axis([0 10.05 0.28 0.68])
grid
hold off

```



5.8 Consider the following training data points for two categories,

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ , which correspond to labels  $y : \{0, 0, 0, 1, 1\}$  respectively.

(a) The first goal is to predict the class label of  $\mathbf{x}_{t1} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$  and  $\mathbf{x}_{t2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  based on the primal TER method utilizing a linear model with a single output (i.e., utilize  $y : \{0, 0, 0, 1, 1\}$  as the single column of learning target).

(b) The second goal is to predict the class label of  $\mathbf{x}_{t1} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$  and  $\mathbf{x}_{t2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  based on the dual TER method utilizing a third-order polynomial model.

5.8 Ans: (a) Primal TER learning utilizing a linear model of single output:  
By packing the data in matrix form:

$$\mathbf{X} = \begin{bmatrix} 1, & 3, & 1 \\ 1, & 2, & 3 \\ 1, & 0, & 1 \\ 1, & 2, & 1 \\ 1, & 1, & 3 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}. \quad (1.1)$$

The TER learning solution for over-determined systems is given by

$$\hat{\mathbf{w}} = \left( \frac{1}{m^+} \mathbf{X}_+^T \mathbf{X}_+ + \frac{1}{m^-} \mathbf{X}_-^T \mathbf{X}_- \right)^{-1} \left( \frac{1}{m^+} \mathbf{X}_+^T \mathbf{y}_+ + \frac{1}{m^-} \mathbf{X}_-^T \mathbf{y}_- \right),$$

where data has been split into the +ve-class and -ve-class matrices as follows:

$$\mathbf{x}_+ = \begin{bmatrix} 1, & 2, & 1 \\ 1, & 1, & 3 \end{bmatrix}, \mathbf{y}_+ = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{x}_- = \begin{bmatrix} 1, & 3, & 1 \\ 1, & 2, & 3 \\ 1, & 0, & 1 \end{bmatrix}, \mathbf{y}_- = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Finally, the prediction can be conducted sample-wise or in multiple samples:

- (i) Single sample:  $\hat{\mathbf{y}}^T = \mathbf{x}_t^T \hat{\mathbf{w}}$  for unseen data sample  $\mathbf{x}_t$  i.e.,  
 $\hat{\mathbf{y}}_1^T = \mathbf{x}_{t1}^T \hat{\mathbf{w}} = 0.5484 > 0.5 \Rightarrow \text{class1},$   
 $\hat{\mathbf{y}}_2^T = \mathbf{x}_{t2}^T \hat{\mathbf{w}} = 0.4516 < 0.5 \Rightarrow \text{class0}$
- (ii) Multiple samples:  $\hat{\mathbf{Y}} = \mathbf{X}_t \hat{\mathbf{w}}$  for unseen data matrix  $\mathbf{X}_t$  packed in similar manner.

(b) Dual TER learning utilizing a third-order polynomial model:

By expanding the data matrix (1.1) into the third-order polynomial  $\mathbf{P}$  matrix for training (consisting of 5 rows/samples of  $\mathbf{p}^T$ ) and the corresponding weight parameter matrix  $\mathbf{W}$ , we have

$$\mathbf{P} = \begin{bmatrix} 1 & x_1 & x_2 & x_1 x_2 & x_1^2 & x_2^2 & x_1^2 x_2 & x_1 x_2^2 & x_1^3 & x_2^3 \\ 1 & 3 & 1 & (3)(1) & 3^2 & 1^2 & 3^2(1) & (3)1^2 & 3^3 & 1^3 \\ 1 & 2 & 3 & (2)(3) & 2^2 & 3^2 & 2^2(3) & (2)3^2 & 2^3 & 3^3 \\ 1 & 0 & 1 & (0)(1) & 0^2 & 1^2 & 0^2(1) & (0)1^2 & 0^3 & 1^3 \\ 1 & 2 & 1 & (2)(1) & 2^2 & 1^2 & 2^2(1) & (2)1^2 & 2^3 & 1^3 \\ 1 & 1 & 3 & (1)(3) & 1^2 & 3^2 & 1^2(3) & (1)3^2 & 1^3 & 3^3 \end{bmatrix}, \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \\ w_6 \\ w_7 \\ w_8 \\ w_9 \end{bmatrix}. \quad (1.2)$$

Next, construct a diagonal matrix  $\mathbf{M}$  using either  $\frac{1}{m^+}$  or  $\frac{1}{m^-}$  at row positions corresponding to their respective category. In other words, each row of  $\mathbf{P}$ ,  $\mathbf{M}$  and  $\mathbf{y}$  should have their binary category labels matched.

- TER learning:  $\hat{\mathbf{w}} = \mathbf{P}^T (\mathbf{MPP}^T)^{-1} \mathbf{M}\mathbf{y}$  for under-determined systems.
- Prediction:

- (i) Single sample:  $\hat{\mathbf{y}}^T = \mathbf{p}(\mathbf{x}_t)^T \hat{\mathbf{w}}$  for unseen data sample  $\mathbf{x}_t$  i.e.,  
 $\hat{\mathbf{y}}_1^T = \mathbf{p}(\mathbf{x}_{t1})^T \hat{\mathbf{w}} = -2.5376 < 0.5 \Rightarrow \text{class0},$   
 $\hat{\mathbf{y}}_2^T = \mathbf{p}(\mathbf{x}_{t2})^T \hat{\mathbf{w}} = 0.6530 > 0.5 \Rightarrow \text{class1}$
- (ii) Multiple samples:  $\hat{\mathbf{Y}} = \mathbf{P}_t \hat{\mathbf{W}}$  for polynomial matrix  $\mathbf{P}_t$  of unseen data packed in similar manner.

5.9 Solve the above question based on one-hot encoding of the target instead of utilizing a single target output.

5.9 Ans: (a) Primal TER learning utilizing multiple target outputs:

By packing the data in matrix form and considering a one-hot encoding for the learning target given by

$$\mathbf{X} = \begin{bmatrix} 1, & 3, & 1 \\ 1, & 2, & 3 \\ 1, & 0, & 1 \\ 1, & 2, & 1 \\ 1, & 1, & 3 \end{bmatrix}, \quad \mathbf{Y} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}. \quad (1.3)$$

The TER learning for over-determined systems is given by

$$\hat{\mathbf{w}}_k = \left( \frac{1}{m^+} \mathbf{X}_+^T \mathbf{X}_+ + \frac{1}{m^-} \mathbf{X}_-^T \mathbf{X}_- \right)^{-1} \left( \frac{1}{m^+} \mathbf{X}_+^T \mathbf{y}_+ + \frac{1}{m^-} \mathbf{X}_-^T \mathbf{y}_- \right),$$

$k = 1, 2$

where data is split into the +ve-class and -ve-class matrices:

$$\mathbf{X}_+ = \begin{bmatrix} 1, & 2, & 1 \\ 1, & 1, & 3 \end{bmatrix}, \mathbf{y}_+ = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{X}_- = \begin{bmatrix} 1, & 3, & 1 \\ 1, & 2, & 3 \\ 1, & 0, & 1 \\ 1, & 2, & 1 \\ 1, & 1, & 3 \end{bmatrix}, \mathbf{y}_- = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \text{ for } k = 1.$$

$$\mathbf{X}_- = \begin{bmatrix} 1, & 2, & 1 \\ 1, & 1, & 3 \end{bmatrix}, \mathbf{y}_- = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ and } \mathbf{X}_+ = \begin{bmatrix} 1, & 3, & 1 \\ 1, & 2, & 3 \\ 1, & 0, & 1 \\ 1, & 2, & 1 \\ 1, & 1, & 3 \end{bmatrix}, \mathbf{y}_+ = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \text{ for } k = 2.$$

Finally, the prediction can be conducted sample-wise or in multiple samples using  $\hat{\mathbf{W}} = [\hat{\mathbf{w}}_1, \hat{\mathbf{w}}_2]$ :

- (i) Single sample:  $\hat{\mathbf{y}}^T = \mathbf{x}_t^T \hat{\mathbf{W}}$  for unseen data sample  $\mathbf{x}_t \Rightarrow$  The column number of the row vector  $\hat{\mathbf{y}}^T$  with the largest value is the class number.  
 $(\hat{\mathbf{y}}_1^T = \mathbf{x}_{t1}^T \hat{\mathbf{W}} = [0.5484, 0.4516], \hat{\mathbf{y}}_2^T = \mathbf{x}_{t2}^T \hat{\mathbf{w}} = [0.4516, 0.5484])$
- (ii) Multiple samples:  $\hat{\mathbf{Y}} = \mathbf{X}_t \hat{\mathbf{W}}$  for unseen data matrix  $\mathbf{X}_t$  packed in similar manner.

(b) Dual TER learning utilizing a third-order polynomial model:

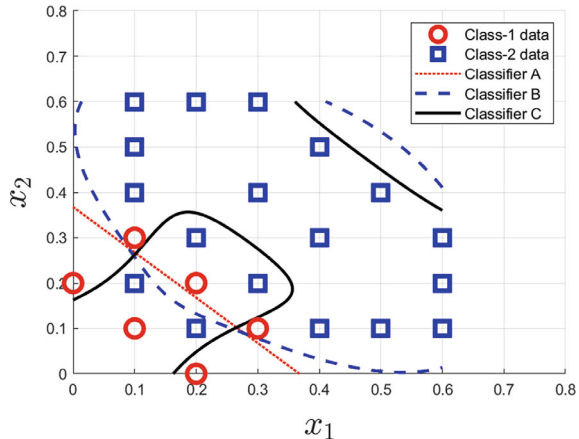
By expanding the data matrix (1.1) into the third-order polynomial  $\mathbf{P}$  matrix for training (consisting of 6 rows/samples of  $\mathbf{p}^T$ ) and the corresponding weight parameter matrix  $\mathbf{W}$ , we have

$$\mathbf{P} = \begin{bmatrix} 1 & x_1 & x_2 & x_1 x_2 & x_1^2 & x_2^2 & x_1^2 x_2 & x_1 x_2^2 & x_1^3 & x_2^3 \\ 1 & 1 & 2 & (1)(2) & 1^2 & 2^2 & 1^2(2) & (1)2^2 & 1^3 & 2^3 \\ 1 & 2 & 0 & (2)(0) & 1^2 & 0^2 & 2^2(0) & (2)0^2 & 2^3 & 0^3 \\ 1 & 1 & 1 & (1)(1) & 1^2 & 1^2 & 1^2(1) & (1)1^2 & 1^3 & 1^3 \\ 1 & 3 & 1 & (3)(1) & 3^2 & 1^2 & 3^2(1) & (3)1^2 & 3^3 & 1^3 \\ 1 & 2 & 3 & (2)(3) & 2^2 & 3^2 & 2^2(3) & (2)3^2 & 2^3 & 3^3 \\ 1 & 3 & 3 & (3)(3) & 3^2 & 3^2 & 3^2(3) & (3)3^2 & 3^3 & 3^3 \end{bmatrix}, \mathbf{W} = \begin{bmatrix} w_{0,1} & w_{0,2} \\ w_{1,1} & w_{1,2} \\ w_{2,1} & w_{2,2} \\ w_{3,1} & w_{3,2} \\ w_{4,1} & w_{4,2} \\ w_{5,1} & w_{5,2} \\ w_{6,1} & w_{6,2} \\ w_{7,1} & w_{7,2} \\ w_{8,1} & w_{8,2} \\ w_{9,1} & w_{9,2} \end{bmatrix}. \quad (1.4)$$

Next, construct a diagonal matrix  $\mathbf{M}_k$  using either  $\frac{1}{m^+}$  or  $\frac{1}{m^-}$  at row positions corresponding to their respective category for each column  $k$ .  $\mathbf{y}_k$ ,  $k = 1, 2$  correspond to the  $k$ -th column of the indicator matrix  $\mathbf{Y}$  in (1.3). In other words, each row of  $\mathbf{P}_k$ ,  $\mathbf{M}_k$  and  $\mathbf{y}_k$  for  $k = 1, 2$  should have their binary category labels matched. Note:  $\mathbf{P}_k$  is the same matrix as  $\mathbf{P}$  but with each +ve and -ve sample aligned with that in  $\mathbf{y}_k$ .

- TER learning:  $\hat{\mathbf{w}}_k = \mathbf{P}_k^T (\mathbf{M}_k \mathbf{P}_k \mathbf{P}_k^T)^{-1} \mathbf{M}_k \mathbf{y}_k$ ,  $k = 1, 2$  for under-determined systems. Pack parameters as  $\hat{\mathbf{W}} = [\hat{\mathbf{w}}_1, \hat{\mathbf{w}}_2]$ .
- Prediction:
  - Single sample:  $\hat{\mathbf{y}}^T = \mathbf{p}(\mathbf{x}_t)^T \hat{\mathbf{W}}$  for unseen data sample  $\mathbf{x}_t \Rightarrow$  The column number of the row vector  $\hat{\mathbf{y}}^T$  with the largest value is the class number.  
 $(\hat{\mathbf{y}}_1^T = \mathbf{p}(\mathbf{x}_{t1})^T \hat{\mathbf{W}} = [-2.5376, 3.7500], \hat{\mathbf{y}}_2^T = \mathbf{p}(\mathbf{x}_{t2})^T \hat{\mathbf{W}} = [0.6530, 0.3232])$
  - Multiple samples:  $\hat{\mathbf{Y}} = \mathbf{P}_t \hat{\mathbf{W}}$  for polynomial matrix  $\mathbf{P}_t$  of unseen data packed in similar manner.

5.10 Given the data points of two categories which are labeled by circles and squares as shown in the following figure. The decision boundaries of Classifier A, Classifier B, and Classifier C are shown as dotted line, dashed line, and solid line respectively.



- (i) Write out the confusion matrix for each classifier.
- (ii) Calculate the Total Error Rate for each Classifier.
- (iii) Which classifier can predict better for unseen data? Why?

5.10 Ans: (i) Classifier A

|    | C1 | C2 |
|----|----|----|
| C1 | 3  | 2  |
| C2 | 3  | 16 |

Classifier B

|    | C1 | C2 |
|----|----|----|
| C1 | 3  | 2  |
| C2 | 3  | 16 |

Classifier C

|    | C1 | C2 |
|----|----|----|
| C1 | 2  | 4  |
| C2 | 4  | 14 |

- (ii) Total Error Rate for each Classifier:

A:  $3/6 + 2/18 = 11/18$

B:  $3/6 + 2/18 = 11/18$

C:  $4/6 + 4/18 = 16/18$

- (iii) Which classifier can predict better for unseen data?

Classifier A and Classifier B show better generalization than Classifier C. They have simpler decision boundary than that of Classifier C.

## Chapter 6

### Penalized Learning

#### Exercises

- 6.1 The solution for a primal ridge regression problem with feature vectors in  $\mathbf{X}$  and targets in  $\mathbf{y}$  can be written as  $\hat{\mathbf{w}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$  for  $\lambda > 0$ . As  $\lambda$  increases,  $\hat{\mathbf{w}}^T \hat{\mathbf{w}}$  increases. True or False?

6.1 Ans: False. Reason: As  $\lambda$  increases, the penalty to  $\hat{\mathbf{w}}^T \hat{\mathbf{w}}$  increases and hence  $\hat{\mathbf{w}}^T \hat{\mathbf{w}}$  decreases.

6.2 Which technique in classifier learning involves minimizing an error cost function while incorporating an  $L_p$ -norm penalty on the model parameters?

- (a) Ridge regression
- (b) Lasso regression
- (c) Elastic net
- (d) Bridge regression
- (e) None of these.

6.2 Ans: (d)

- (a) Ridge regression: incorporating an  $L_2$ -norm penalty on the model parameters.
- (b) Lasso regression: incorporating an  $L_1$ -norm penalty on the model parameters.
- (c) Elastic net: varies between  $L_1$ -norm and  $L_2$ -norm penalty on the model parameters.
- (d) Bridge regression: incorporating an  $L_p$ -norm penalty on the model parameters.
- (e) None of these.

6.3 Consider the penalized learning formulated by

$$\min_{\mathbf{w}} J(\mathbf{x}, \mathbf{w}) = \min_{\mathbf{w}} \left\{ \sum_{i=1}^m (f(\mathbf{x}_i, \mathbf{w}) - y_i)^2 + \lambda \|\mathbf{w}\|_p^p \right\}, \quad \lambda > 0$$

where  $f(\mathbf{x}, \mathbf{w})$  is a learning model with augmented input  $\mathbf{x} \in \mathbb{R}^{d+1}$ , learning parameter  $\mathbf{w} \in \mathbb{R}^{d+1}$  and  $y_i$ ,  $i = 1, \dots, m$  being the learning target. Which of the following statement(s) is/are true?

- (a) When  $p \leq 1$ , it performs both variable selection and regularization.
- (b) When  $p > 2$ , it does not perform coefficient shrinkage.
- (c) When  $p < 2$ , it performs coefficient shrinkage.
- (d) When  $p = 2$ , the solution is sparse.
- (e) None of these.

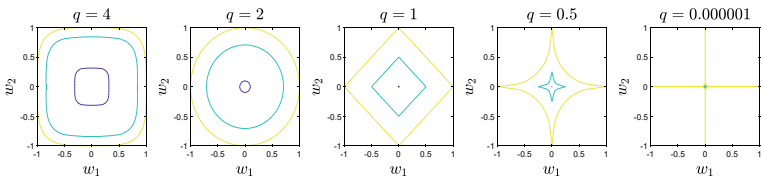
6.3 Ans: (a) and (c)

Reason: the method performs coefficient shrinkage for all values of  $p$  since the coefficient norm term is also minimized along minimization of the error term when  $\lambda > 0$ .

6.4 Consider the criterion

$$\hat{\mathbf{w}} = \min_{\mathbf{w}} \left\{ \sum_{i=1}^m \left( y_i - w_0 - \sum_{j=1}^d x_{ij} w_j \right)^2 + \lambda \sum_{j=1}^d |w_j|^q \right\}, \quad q \geq 0,$$

where  $x$  and  $y$  are the input and target values,  $\boldsymbol{w}$  is the parameter vector and  $\lambda$  is a penalty factor. The contours of constant value of  $\sum_j |\boldsymbol{w}_j|^q$  are shown in figure below for the case of two inputs. Write down the common name/term (for those items marked with “?”) for each value of  $q$  and state whether there is coefficient shrinkage and sparseness for each estimation (by circling the answer) (Note: the answer for LASSO is already provided):



| Value of $q$ | Common Names | Coefficient shrinkage? | Sparse?   |
|--------------|--------------|------------------------|-----------|
| $q = 0$      | ?            | (Yes/No)?              | (Yes/No)? |
| $q = 0.5$    | –NA–         | (Yes/No)?              | (Yes/No)? |
| $q = 1$      | lasso        | Yes                    | Yes       |
| $q = 2$      | ?            | (Yes/No)?              | (Yes/No)? |
| $q = 4$      | –NA–         | (Yes/No)?              | (Yes/No)? |

| Value of $q$ | Common Terms              | Coefficient shrinkage? | Sparse? |
|--------------|---------------------------|------------------------|---------|
| $q = 0$      | variable subset selection | (Yes)                  | (Yes)   |
| $q = 0.5$    | –                         | (Yes)                  | (Yes)   |
| $q = 1$      | lasso                     | (Yes)                  | (Yes)   |
| $q = 2$      | ridge regression          | (Yes)                  | (No)    |
| $q = 4$      | –                         | (Yes)                  | (No)    |

- 6.4 Ans:
- 6.5 Consider the XOR problem with four training data samples given by  $(x_1, x_2) \in \{(0, 1), (2, 1), (1, 0), (1, 2)\}$  with their corresponding target outputs  $y \in \{0, 0, 1, 1\}$ . Use the following quadratic polynomial model to learn the XOR data based on the proximal bridge regression at  $k = 1.1$  and  $k = 2$  using  $\lambda = 10$ .

$$p(x_1, x_2) = w_0 + w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_2^2 + w_5x_1x_2$$

Rank the coefficients in terms of their magnitudes. Comment on any difference in the rankings.

- 6.5 Ans: This is an under-determined case. The MSEs at  $k = 1.1$  and  $k = 2$  are respectively 0.4478 and 0.3295. This shows compromise of fitting accuracy to favor sparseness for  $k = 1.1$ .  
The estimated parameter vectors are respectively

$$\mathbf{w}_{k=1.1} = \begin{bmatrix} 0.0001 \\ 0.0005 \\ -0.0005 \\ -0.0416 \\ 0.0516 \\ 0.0018 \end{bmatrix}, \quad \mathbf{w}_{k=2} = \begin{bmatrix} 0.0278 \\ 0.0213 \\ 0.0366 \\ -0.0223 \\ 0.0656 \\ 0.0300 \end{bmatrix}.$$

The ranking orders for the learned coefficients are

$$\mathbf{w}_{k=1.1} : w_4, w_3, w_5, w_1/w_2, w_0,$$

$$\mathbf{w}_{k=2.0} : w_4, w_2, w_5, w_0, w_3, w_1.$$

The above magnitude ranking shows the different importance of the coefficients between the learned parameters of  $\mathbf{w}_{k=1.1}$  and  $\mathbf{w}_{k=2}$ . The coefficient  $w_4$  for  $x_2^2$  in both cases shows the highest influence in determining the learning decision.

Codes in MATLAB:

```
% XOR training data
X = [ 0  1; 2  1; 1  0; 1  2]; % XOR inputs (training data)
ytr = [0; 0; 1; 1];           % XOR outputs (training data)

%--- 2nd-order Polynomial features ---%
Ptr = [ones(4,1), X(:,1), X(:,2), X(:,1).^2, X(:,2).^2, X(:,1).*X(:,2)];
Lamval = [10];
kval = [1.1, 2];
for nn = 1:length(Lamval)
    for ll=1:length(kval)
        alpha(:,nn,ll) = p_bridge(Ptr,ytr,kval(ll),Lamval(nn));
        ytr_bridge = Ptr*alpha(:,nn,ll);
        MSE(nn,ll) = mean((ytr - ytr_bridge).^2);
    end
end

function [alpha] = p_bridge(P,y,k,lambdap,primal)
% primal = "1", "0" for primal, dual respectively
% exclusion of "primal" -> default choice based on over- or under-determined system
if (nargin==4) primal = 2; end % default choice if no input for primal
if (primal==1) Prow = 2; Pcol = 1;
elseif (primal==0) Prow = 1; Pcol = 2;
else [Prow,Pcol] = size(P); end
if (Prow<Pcol) % under-determined case
    if (k==2)
        Pwarp = P';
    else
        Pwarp = abs(P').^(1/(k-1));
    end
end
```

```

ID = eye(size(P*P'));
III = (P'/(P*P' + lambda*ID))*P;
VVV = (Pwarp/(P*Pwarp + lambda*ID))*y;
alpha = sign(VVV).*abs( (III*(VVV.^(k-1))).^(1/(k-1)) );
else % over-determined case
pseudoinv = (P'*P + lambda*eye(size(P'*P)))\P';
alpha = pseudoinv*y; % initial estimation
for kk=1:5
    pseudoinv = (diag((lambda*k/2)*(abs(alpha(:,1))).^(k-2))+P'*P)\P';
    alpha = pseudoinv*y;
end
end
end % end function

```

6.6 Consider the simulated data with the true model given by  $y = \mathbf{w}^T \mathbf{x} + \sigma \epsilon$  with  $\epsilon \sim N(0, 1)$ . Generate a data set consisting of 100/100/400 (train-validation/test) samples with  $\sigma = 15$  and

$$\mathbf{w} = [0, 3, 6, 8, 0, 0, 0, 0, 0]^T.$$

The correlation between  $x_i$  and  $x_j$  is zero.

(Hint: the data can be generated using the following MATLAB code:

```
Xall = mvnrnd(zeros(1,10),diag(ones(1,10)),600);
```

which uses a fixed random seed such as `rng(8)`.)

Compute the Mean-Squared Error (MSE) of learning using ordinary least-squares, ridge regression and proximal bridge regression based on the generated data. Compare their learned parameters too.

(Hint: since the true model is known, the MSE can be computed using  $\text{MSE} = (\hat{\mathbf{w}} - \mathbf{w})^T \frac{\mathbf{X}_t^T \mathbf{X}_t}{n} (\hat{\mathbf{w}} - \mathbf{w})$  where  $\mathbf{X}_t$  and  $n$  are respectively the matrix containing the test samples and the test sample size.)

```

sig = 15; beta = [0,3,6,8,0,0,0,0,0,0]';
for kk = 1:1 %for a single trial
    %--- Generate data ---%
    rng(kk*8); Xall = mvnrnd(zeros(1,10),diag(ones(1,10)),600);
    %-----%
    [nrow,ncol,ntrial] = size(Xall);
    X = Xall; % no bias term
    rng(kk*18); epsilon = normrnd(0,1,600,1);
    y = X*beta + sig*epsilon; % target, exclude bias term
    rng(kk*68); Idx = randperm(600);
    Xrand = X(Idx,:); yrand = y(Idx);
    Xtr = Xrand(1:100,:); ytr = yrand(1:100);
    Xva = Xrand(101:200,:); yva = yrand(101:200);
    Xts = Xrand(201:600,:); yts = yrand(201:600);

```

```

%--- OLS regression ---%

alpha_ols = ridge(ytr,Xtr,0);

yts_ols = Xts*alpha_ols;

MSE_ols(kk) = (alpha_ols - beta)'*(Xts'*Xts)*(alpha_ols - beta)/length(yts);

ALPHA_ols(:,kk) = alpha_ols;

%--- ridge regression ---%

clear Lam; clear kval; clear minVal; clear minIdx;

Lam = [0:.01:1,2:1:10,20:10:100,200:100:1000,2000:1000:10000];

for jj=1:length(Lam)

    alpha_ridge = ridge(ytr,Xtr,Lam(jj));

    yva_ridge = Xva*alpha_ridge;

    MSE_ridgeval(jj) = mean((yva - yva_ridge).^2);

end

[minVal,minIdx] = min(MSE_ridgeval);

alpha_ridge = ridge(ytr,Xtr,Lam(minIdx));

yts_ridge = Xts*alpha_ridge;

MSE_ridge(kk) = (alpha_ridge - beta)'*(Xts'*Xts)*(alpha_ridge - beta)/length(yts);

ALPHA_ridge(:,kk) = alpha_ridge;

Lam_ridge = Lam(minIdx);

%--- pbridge regression k=min ---%

clear Lam; clear kval; clear minVal; clear minIdx; clear MSE_bridgeval;

Lam = [0:.01:1,2:1:10,20:10:100,200:100:1000,2000:1000:10000];

kval = [1:.01:2];

clear alpha_bridge; clear yva_bridge; clear MSE_bridgeval;

for jj=1:length(Lam)

    [kk,2,jj]

    for ll=1:length(kval)

        alpha_bridge = bridge(Xtr,ytr,kval(ll),Lam(jj));

        yva_bridge = Xva*alpha_bridge;

        MSE_bridgeval(jj,ll) = mean((yva - yva_bridge).^2);

    end

end

[minVal1,minIdx1] = min(MSE_bridgeval,[],1);

[minVal2,minIdx2] = min(minVal1);

alpha_bridgekmin = bridge(Xtr,ytr,kval(minIdx2),Lam(minIdx1(minIdx2)));

yts_bridgekmin = Xts*alpha_bridgekmin;

MSE_bridgekmin(kk) = (alpha_bridgekmin - beta)'*(Xts'*Xts)*(alpha_bridgekmin - beta)/length(yts);

ALPHA_bridgekmin(:,kk) = alpha_bridgekmin;

k_bridge = kval(minIdx2);

Lam_bridge = Lam(minIdx1(minIdx2));

end

median_MSE_ols = median(MSE_ols);

```

```

std_MSE_ols = std(MSE_ols);
OLS_result = [median_MSE_ols, std_MSE_ols/sqrt(length(MSE_ols))]
median_MSE_ridge = median(MSE_ridge);
std_MSE_ridge = std(MSE_ridge);
Ridge_result = [median_MSE_ridge, std_MSE_ridge/sqrt(length(MSE_ridge))]
median_MSE_bridgekmin = median(MSE_bridgekmin);
std_MSE_bridgekmin = std(MSE_bridgekmin);
Bridgekmin_result = [median_MSE_bridgekmin, std_MSE_bridgekmin/sqrt(length(MSE_bridgekmin))]

% display learned parameters
[mean(ALPHA_ols,2), mean(ALPHA_ridge,2), mean(ALPHA_bridgekmin,2)]

```

The selected  $k$  and  $\lambda$  values for proximal bridge are respectively 1.01 and 300. The selected  $\lambda$  for ridge regression is 40. The learned parameters are

$$[\hat{\mathbf{w}}_{\text{ols}}, \hat{\mathbf{w}}_{\text{ridge}}, \hat{\mathbf{w}}_{\text{pbridge}}] = \begin{bmatrix} -0.6719 & 0.0770 & 0.0002 \\ 5.0816 & 3.2727 & 3.1247 \\ 7.2959 & 5.2663 & 5.9396 \\ 9.6343 & 6.7390 & 7.8913 \\ 0.8932 & 0.5700 & 0.0073 \\ 0.4527 & -0.4748 & -0.0009 \\ -3.6663 & -2.4003 & -1.5043 \\ -1.6277 & -0.6048 & -0.0005 \\ -0.6747 & -0.7199 & -0.0127 \\ -0.8095 & -0.7749 & -0.0370 \end{bmatrix}.$$

The MSE values for  $\hat{\mathbf{w}}_{\text{ols}}$ ,  $\hat{\mathbf{w}}_{\text{ridge}}$ ,  $\hat{\mathbf{w}}_{\text{pbridge}}$  are respectively 25.9375, 9.9451, 2.2504. These results show the compressive capability of proximal bridge.

6.7 For the over-determined and under-determined solutions of the proximal bridge regression, are the estimations biased?

6.7 Ans: We shall work on the single output case only since the multiple outputs case is a direct stacking of the single output case. Assume the data is generated according to  $\mathbf{y} = \mathbf{X}\boldsymbol{\alpha} + \boldsymbol{\epsilon}$  with  $\mathbf{X} \in \mathbb{R}^{M \times D}$ ,  $\boldsymbol{\alpha} \in \mathbb{R}^D$  where  $\boldsymbol{\epsilon}$  is a zero mean noise with covariance matrix  $\mathbf{C}$ . For the over-determined case, we have  $M \geq D$ . Suppose the estimation is initialized by  $\boldsymbol{\alpha}_0$ , then the expectation of  $\hat{\boldsymbol{\alpha}}$  is

$$\begin{aligned}
E[\hat{\boldsymbol{\alpha}}] &= E \left[ \left( \frac{\lambda k}{2} \text{diag}\{|\boldsymbol{\alpha}_0|^{\circ(k-2)}\} + \mathbf{X}^T \mathbf{X} \right)^{-1} \mathbf{X}^T (\mathbf{X}\boldsymbol{\alpha} + \boldsymbol{\epsilon}) \right] \\
&= \left( \frac{\lambda k}{2} \text{diag}\{|\boldsymbol{\alpha}_0|^{\circ(k-2)}\} + \mathbf{X}^T \mathbf{X} \right)^{-1} \mathbf{X}^T \mathbf{X} \boldsymbol{\alpha} \\
&\neq \boldsymbol{\alpha}, \quad \forall k > 1, \lambda > 0.
\end{aligned} \tag{1.5}$$

This shows that the estimation is biased for the ranges of our working  $k$  values and  $\lambda$  values. For the special case when  $\lambda = 0$ , we have an unbiased ols estimation since  $E[\hat{\alpha}] = E\left[(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T (\mathbf{X}\alpha + \epsilon)\right] = E\left[(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X}\alpha\right] = \alpha$ .

For the under-determined system where  $M < D$ , the expectation of estimate is

$$\begin{aligned} E[\hat{\alpha}] &= E\left[\text{sgn}(\theta) \circ \left|\mathbf{X}^T (\mathbf{X}\mathbf{X}^T)^{-1} \mathbf{X}\theta^{\circ(k-1)}\right|^{\circ \frac{1}{k-1}}\right], \\ \theta &= |\mathbf{X}^T|^{\circ \frac{1}{k-1}} \left[\mathbf{X}|\mathbf{X}^T|^{\circ \frac{1}{k-1}}\right]^{-1} (\mathbf{X}\alpha + \epsilon), \\ &\neq \alpha. \end{aligned} \quad (1.6)$$

The inequality holds because the rank of

$$\left|\mathbf{X}^T (\mathbf{X}\mathbf{X}^T)^{-1} \mathbf{X} \left\{|\mathbf{X}^T|^{\circ \frac{1}{k-1}} \left[\mathbf{X}|\mathbf{X}^T|^{\circ \frac{1}{k-1}}\right]^{-1} \mathbf{X}\alpha\right\}^{\circ(k-1)}\right|^{\circ \frac{1}{k-1}}$$

is at most  $M$  (due to  $(\mathbf{X}\mathbf{X}^T)^{-1}$ ) which is smaller than  $D$ . The above analyses for the over- and the under-determined cases show that both the estimates are biased, and this is consistent with the compressed estimation where some ideal parameters have been suppressed.

6.8 The penalty term in ridge regression can be considered as an augmentation data set on the ordinary least squares. By augmenting the data matrix  $\mathbf{X}$  with  $d$  additional rows and the target  $\mathbf{y}$  with  $d$  zeros, show that such relationship between data augmentation and regularization can be established.

6.8 Ans: Consider  $\mathbf{X} \in \mathbb{R}^{d \times d}$  with corresponding target  $\mathbf{y} \in \mathbb{R}^{d \times 1}$ . By appending the identity matrix  $\mathbf{I}$  to  $\mathbf{X}$  with appropriate scaling and appending zeros  $\mathbf{0}$  to  $\mathbf{y}$ , we have

$$\tilde{\mathbf{X}} = \begin{bmatrix} \mathbf{X} \\ \sqrt{\lambda} \mathbf{I} \end{bmatrix}, \quad \tilde{\mathbf{y}} = \begin{bmatrix} \mathbf{y} \\ \mathbf{0} \end{bmatrix}.$$

The least squares solution to this augmented problem is

$$\mathbf{w} = (\tilde{\mathbf{X}}^T \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^T \tilde{\mathbf{y}}$$

where the matrices can be expanded in terms of their composing blocks:

$$\begin{aligned} \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} &= [\mathbf{X}^T \quad \sqrt{\lambda} \mathbf{I}] \begin{bmatrix} \mathbf{X} \\ \sqrt{\lambda} \mathbf{I} \end{bmatrix} = \mathbf{X}^T \mathbf{X} + \lambda \mathbf{I}, \\ \tilde{\mathbf{X}}^T \tilde{\mathbf{y}} &= [\mathbf{X}^T \quad \sqrt{\lambda} \mathbf{I}] \begin{bmatrix} \mathbf{y} \\ \mathbf{0} \end{bmatrix} = \mathbf{X}^T \mathbf{y}. \end{aligned}$$

This solution to the augmented problem can thus be written as

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}.$$

6.9 By utilizing the singular value decomposition of the centered regressor matrix  $\mathbf{X}$ , derive an expression to show that  $\|\mathbf{w}\|$  increases as its tuning parameter  $\lambda \rightarrow 0$ .

6.9 Ans: The singular value decomposition of the centered regressor matrix  $\mathbf{X} \in \mathbb{R}^{m \times d}$  can be writtend as  $\mathbf{X} = \mathbf{U} \mathbf{D} \mathbf{V}^T$  where  $\mathbf{U}$  is a  $m \times d$  matrix with orthonormal columns that span the column space of  $\mathbf{X}$ , and  $\mathbf{V}$  is a  $d \times d$  orthogonal matrix, and finally  $\mathbf{D}$  is a  $d \times d$  diagonal matrix with elements  $d_j$  ordered such that  $d_1 \geq d_2 \geq \dots \geq d_d \geq 0$ . Based on this decomposition, we have

$$\mathbf{X}^T \mathbf{X} = \mathbf{V} \mathbf{D} \mathbf{U}^T \mathbf{U} \mathbf{D} \mathbf{V}^T = \mathbf{V} \mathbf{D}^2 \mathbf{V}^T.$$

Based on the above, we have

$$\begin{aligned} \mathbf{w} &= (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y} \\ &= (\mathbf{V} \mathbf{D}^2 \mathbf{V}^T + \lambda \mathbf{V} \mathbf{V}^T)^{-1} \mathbf{V} \mathbf{D} \mathbf{U}^T \mathbf{y} \\ &= (\mathbf{V} (\mathbf{D}^2 + \lambda \mathbf{I}) \mathbf{V}^T)^{-1} \mathbf{V} \mathbf{D} \mathbf{U}^T \mathbf{y} \\ &= (\mathbf{V} (\mathbf{D}^2 + \lambda \mathbf{I}))^{-1} \mathbf{D} \mathbf{U}^T \mathbf{y}. \end{aligned}$$

Utilizing this result, we then have

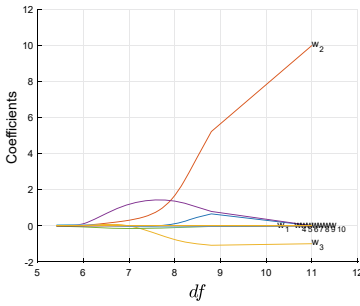
$$\begin{aligned} \|\mathbf{w}\|_2^2 &= \mathbf{y}^T \mathbf{U} \mathbf{D} (\mathbf{D}^2 + \lambda \mathbf{I})^{-1} \mathbf{V}^T \mathbf{V} (\mathbf{D}^2 + \lambda \mathbf{I})^{-1} \mathbf{D} \mathbf{U}^T \mathbf{y} \\ &= \mathbf{y}^T \mathbf{U} \mathbf{D} (\mathbf{D}^2 + \lambda \mathbf{I})^{-2} \mathbf{D} \mathbf{U}^T \mathbf{y} \\ &= (\mathbf{U}^T \mathbf{y})^T [\mathbf{D} (\mathbf{D}^2 + \lambda \mathbf{I})^{-2} \mathbf{D}] (\mathbf{U}^T \mathbf{y}). \end{aligned}$$

Since the middle matrix is diagonal with elements  $\frac{d_j^2}{(d_j^2 + \lambda)^2}$ , we thus have

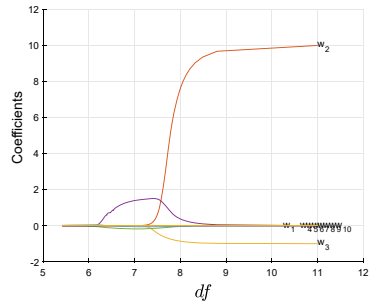
$$\|\mathbf{w}\|_2^2 = \sum_{j=1}^d \frac{d_j^2 (\mathbf{U}^T \mathbf{y})_j^2}{(d_j^2 + \lambda)^2},$$

with  $(\mathbf{U}^T \mathbf{y})_j$  indicating the  $j$ th component of the vector  $(\mathbf{U}^T \mathbf{y})$ . Therefore, as  $\lambda \rightarrow 0$ , the fraction  $\frac{d_j^2}{(d_j^2 + \lambda)^2}$  increases.

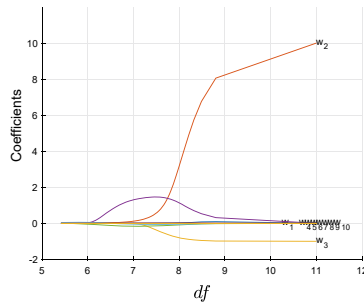
6.10 Match among the types ( $\ell_1$ -norm,  $\ell_2$ -norm,  $\ell_{1.5}$ -norm) of penalized learning with their corresponding parameter profile below:



(a)



(b)



(c)

- 6.10 Ans: (a)  $\ell_2$ -norm: the most gentle compression of parameters  
 (b)  $\ell_1$ -norm: the steepest compression of parameters with some go to zero  
 (c)  $\ell_{1.5}$ -norm: moderate compression

## Chapter 7

### Network Learning

#### Exercises

- 7.1 Solving the parameters of the system of linear equations under the kernel and the range spaces result in the minimum  $\ell_2$ -norm of the parameters. True or False?

7.1 Ans: True. See Lemma 7.1.

- 7.2 Utilization of the  $\ell_1$  loss in network residual learning leads to sparseness of the network weights. True or False?

7.2 Ans: False. This is because the learning residual will be forced to be sparse but not the weights.

7.3 Select from below any ways that are effective to reduce overfitting in neural networks.

- (a) Including an  $L_2$  regularization.
- (b) Increasing the number of hidden nodes.
- (c) Increasing the learning rate.
- (d) Augmenting the training data with synthetic samples.

7.3 Ans: (a) and (d).

7.4 Consider two networks using the same activation function in all their layers. Then, when trained appropriately, a neural network with 3 layers will not have a higher training error than that of a single-layer neural network.

7.4 Ans: True. This is because a single-layer network does not have the universal approximation capability possessed by a 3-layer network.

7.5 Suppose you have trained a network for a huge number of iterations with high training and test errors. Which of the following, done in isolation, has a better chance of reducing both the training and test errors?

- (a) Add more hidden layer(s).
- (b) Add more nodes to hidden layers.
- (c) Add more training data.
- (d) Reduce connections of hidden layer(s).
- (e) None of these.

7.5 Ans: (a) Add more hidden layer(s): this might not improve the situation when data is not increased.

(b) Add more nodes to hidden layers: tends to overfit when overly trained.

(c) **Add more training data: both training and test errors decrease if added data is representative.**

(d) Reduce connections of hidden layer(s): this might increase the training error while decreasing the test error marginally.

7.6 Consider a three-layer neural network utilizing the sigmoid function given by  $1/(1 + \exp(-\beta x))$  as the activation in all layers. The parameter  $\beta$  of the sigmoid controls the slope of the activation function. A training error of 89.35 and a testing error of 91.8 are observed. Which of the following is true?

- (a) Increasing  $\beta$  will most likely increase the test error.
- (b) Increasing  $\beta$  will most likely reduce the test error.
- (c) Decreasing  $\beta$  will most likely reduce the test error.
- (d) Decreasing  $\beta$  will most likely increase the test error.
- (e) Not enough information is provided to determine how  $\beta$  should be changed.
- (f) None of these.

7.6 Ans: (e)

7.7 A fully connected network of 3 layers can be written as

$$\mathbf{G} = f([\mathbf{1}, f([\mathbf{1}, f(\mathbf{X}\mathbf{W}_1)]\mathbf{W}_2)]\mathbf{W}_3)$$

where

$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 3.0 \\ 1 & 2 & 2.5 \end{bmatrix}, \quad \mathbf{W}_1 = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix},$$

$$\mathbf{W}_2 = \mathbf{W}_3 = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix}.$$

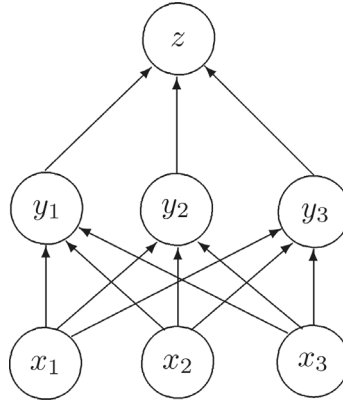
Suppose the Rectified Linear Unit (ReLU) has been used as the activation function ( $f$ ) for all the nodes. Compute the network output matrix  $\mathbf{G}$  based on the given network weights and data.

7.7 Ans: Utilizing MATLAB to compute the network output, we have

```
x = [1 3; 2 2.5];
X = [ones(2,1), x];
W1 = [-1 0 1; 0 -1 0; 1 0 1];
W2 = [-1 0 1; 0 -1 0; 1 0 1; 1 -1 1];
W3 = W2;
F = ReLU([ones(2,1), ReLU([ones(2,1), ReLU(X*W1)]
*W2)]*W3)
*[2mm] function g = ReLU(a)
g = max(a, 0);
end
```

The network output is  $\begin{bmatrix} 4.0 & 0 & 6.0 \\ 3.5 & 0 & 5.5 \end{bmatrix}$ .

7.8 The following graph shows the structure of a simple neural network with a single hidden layer. The input layer consists of three dimensions  $\mathbf{x} = [x_1, x_2, x_3]^T$ . The hidden layer includes three units  $\mathbf{y} = [y_1, y_2, y_3]^T$ . The output layer includes one unit  $z$ . The bias terms are ignored for simplicity.



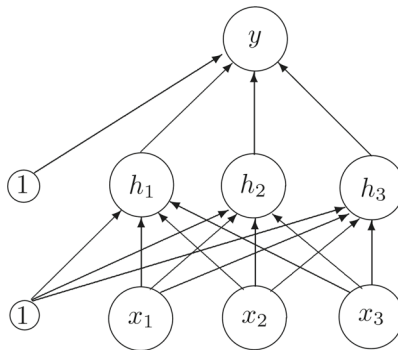
A linear rectified unit  $\text{ReLU}(a) = \max(0, a)$  is adopted as activation function for the hidden layer and the output layer. Given the loss function  $L(z, t) = \frac{1}{2}(z - t)^2$  with  $t$  being the target value for learning. Denote by  $\mathbf{W}_1$  and  $\mathbf{W}_2$  weight matrices connecting input and hidden layer, and hidden layer and output respectively. They are initialized as follows:

$$\mathbf{W}_1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad \mathbf{W}_2 = \begin{bmatrix} 1 & -1 & 1 \end{bmatrix}.$$

Write out symbolically (no need to put any specific values of  $\mathbf{W}_1$  and  $\mathbf{W}_2$ ) the feedforward functional mapping  $\mathbf{x} \rightarrow z$  using  $\text{ReLU}$ ,  $\mathbf{W}_1$ ,  $\mathbf{W}_2$ . In other words, express  $y$  in terms of  $\text{ReLU}$ ,  $\mathbf{W}_1$ ,  $\mathbf{W}_2$  and  $\mathbf{x}$ .

7.8 Ans:  $z = \text{ReLU}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{x}))$  or  $z = \text{ReLU}(\text{ReLU}(\mathbf{x}^T \mathbf{W}_1^T) \mathbf{W}_2^T)$ .

7.9 Consider the network  $N$  below. The input layer of the network consists of three dimensions  $\mathbf{x} = [x_1, x_2, x_3]^T$  and the bias. The hidden layer includes three units  $\mathbf{h} = [h_1, h_2, h_3]^T$  and the bias. The output layer includes one unit  $y$ .



After training this network, the values of the weights (including the bias weights) are

$$\mathbf{W}_1 = \begin{bmatrix} 1 & -1 & 0.5 & 1 \\ 0.5 & -1 & 0.2 & -1 \\ 1 & -0.5 & 1 & 1 \end{bmatrix}, \quad \mathbf{w}_2 = [1 \ 0.5 \ 1 \ -1]^T.$$

(a) Suppose the same activation function  $\sigma$  has been adopted for all the hidden and output nodes. Write out symbolically (no need to plug in the specific values of  $\mathbf{W}_1$  and  $\mathbf{w}_2$  just yet) the feedforward functional mapping  $N: \mathbf{x} \mapsto y$  using  $\sigma$ ,  $\mathbf{W}_1$ ,  $\mathbf{w}_2$ . In other words, express  $y$  in terms of  $\sigma$ ,  $\mathbf{W}_1$ ,  $\mathbf{w}_2$  and  $\mathbf{x}$ .

(b) Write down the gradient of the loss function with respect to the weights. In other words, compute the following terms symbolically:

- (i) The gradient with respect to  $\mathbf{w}_2$ .
- (ii) The gradient with respect to  $\mathbf{W}_1$ .

(c) Suppose you input a three-dimensional sample  $\mathbf{x} = [1, 3, 1]^T$  to the above network  $N$ . You tried four different types of activation functions  $(\phi_1, \phi_2, \phi_3, \phi_4)$  (on all the hidden and output nodes) which respectively output the values  $(y_1, y_2, y_3, y_4) = (0, 0.6818, -3.1500, -0.9963)$ . What is a valid guess for the activation functions  $(\phi_1, \phi_2, \phi_3, \phi_4)$ ?

7.9 Ans: (a)

$$\begin{aligned} \mathbf{h} &= \sigma(\mathbf{W}_1 \mathbf{x}) \\ y &= \sigma(\mathbf{w}_2^T \mathbf{h}) \end{aligned}$$

(b)

- (i) The gradient relative to  $\mathbf{w}_2$ .

$$\frac{\partial L}{\partial \mathbf{w}_2} = \left( \frac{\partial y}{\partial \mathbf{w}_2} \right)^T \frac{\partial L}{\partial y}$$

- (ii) The gradient relative to  $\mathbf{W}_1$ .

$$\frac{\partial L}{\partial \mathbf{W}_1} = \left( \frac{\partial \mathbf{h}}{\partial \mathbf{W}_1} \right)^T \left( \frac{\partial y}{\partial \mathbf{h}} \right)^T \frac{\partial L}{\partial y}$$

(c) MATLAB codes:

```

 $\phi_1$ : function output = ReLU(x)
    output = max(0,x);
end

 $\phi_2$ : function output = sig(x)
    output = 1./(1+exp(-x));
end

 $\phi_3$ : function output = lin(x)
    output = x;
end

 $\phi_4$ : tanh()

```

7.10 Consider a binary classification task of classifying images as apple versus non-apple. You design a deep network with a single output neuron  $z$  for the learning. To make the binary decision, this output neuron goes through an activation given by  $\hat{y} = \sigma(\text{ReLU}(z))$  where  $\sigma(\cdot)$  denotes the sigmoid function and  $\text{ReLU}(\cdot)$  denotes the linear rectified unit. Identify all correct setting(s) below to get full credit.

- (a) We can set the target as  $y \in \{-1, +1\}$  for apple and non-apple respectively and then classify the output as apple when  $\hat{y} \geq 0$  and non-apple when  $\hat{y} < 0$ .
- (b) We can set the target as  $y \in \{0, 1\}$  for apple and non-apple respectively and then classify the output as apple when  $\hat{y} \geq 0.5$  and non-apple when  $\hat{y} < 0.5$ .
- (c) We can set the target as  $y \in \{-1, +1\}$  for non-apple and apple respectively and then classify the output as non-apple when  $\hat{y} \geq 0$  and apple when  $\hat{y} < 0$ .
- (d) We can set the target as  $y \in \{0, 1\}$  for non-apple and apple respectively and then classify the output as non-apple when  $\hat{y} \geq 0.5$  and apple when  $\hat{y} < 0.5$ .
- (e) None of these

7.10 Ans: (e) None of these

(Since the output of ReLU is always positive, the output of sigmoid is always  $\sigma(\text{ReLU}(z)) \geq 0.5$ . Hence it is impossible to make decision based on the listed settings )

## Chapter 8

### Ensemble Learning

#### Exercises

8.1 Explain how the correlation between base learners in a bagging ensemble affects the overall variance of the ensemble prediction.

8.1 Ans: The variance of the ensemble prediction is affected by the correlations between the predictions of individual base learners. To understand this, consider the ensemble prediction  $F(\mathbf{x})$  which is the average of predictions from  $M$  base learners  $h_m(\mathbf{x})$ , i.e.,

$$F(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M h_m(\mathbf{x}).$$

Assuming each learner has the same variance  $\sigma^2$  and the pairwise correlation between different learners' predictions is  $\rho$ , the overall variance of the ensemble's prediction can be derived as follows:

$$\text{Var}(F(\mathbf{x})) = \text{Var}\left(\frac{1}{M} \sum_{m=1}^M h_m(\mathbf{x})\right) = \frac{1}{M^2} \text{Var}\left(\sum_{m=1}^M h_m(\mathbf{x})\right).$$

Given that the variance of the sum of correlated variables is

$$\text{Var}\left(\sum_{m=1}^M h_m(\mathbf{x})\right) = \sum_{m=1}^M \text{Var}(h_m(\mathbf{x})) + \sum_{m \neq n} \text{Cov}(h_m(\mathbf{x}), h_n(\mathbf{x})),$$

where  $\text{Var}(h_m(\mathbf{x})) = \sigma^2$  for all  $m$  and  $\text{Cov}(h_m(\mathbf{x}), h_n(\mathbf{x})) = \rho\sigma^2$ , the equation becomes

$$\text{Var}\left(\sum_{m=1}^M h_m(\mathbf{x})\right) = M\sigma^2 + M(M-1)\rho\sigma^2.$$

Substituting back, we get

$$\text{Var}(F(\mathbf{x})) = \frac{\sigma^2}{M} + \frac{M-1}{M} \rho\sigma^2.$$

Rearranging, this yields

$$\text{Var}(F(\mathbf{x})) = \rho\sigma^2 + \frac{1-\rho}{M}\sigma^2.$$

This formulation shows that the variance of the ensemble decreases as the number of learners  $M$  increases, but the reduction is limited by the correlation  $\rho$ . If learners are completely uncorrelated ( $\rho = 0$ ), variance reduction is maximal. Conversely, if learners are fully correlated ( $\rho = 1$ ), multiple learners do not reduce variance.

8.2 Explain the concept of “Irreducible Error” and how it impacts the performance of ensemble learning methods like bagging and boosting.

8.2 Ans: Irreducible error refers to the error inherent in the problem itself, due to noise or other data-related issues that cannot be eliminated by any model. This type of error affects all predictive models, including ensemble methods. While bagging and boosting can significantly reduce variance and bias, respectively, they cannot affect the irreducible error. Understanding this limitation is crucial for realistic expectations of model performance and emphasizes the importance of high-quality, clean data for training predictive models.

8.3 Discuss the importance of “Feature Importance” in the context of Random Forest and how it is calculated.

8.3 Ans: Feature importance in Random Forest is crucial for understanding the influence of each feature on the prediction. It is calculated based on the decrease in node impurity (e.g., Gini impurity for classification tasks) that results from splits on each feature. Averaging this decrease across all trees in the forest gives a measure of the overall importance of each feature. This metric not only helps in model interpretation but also guides feature selection and engineering, allowing for more focused model optimization.

8.4 Describe the effect of “Learning Rate” in boosting algorithms like AdaBoost and XGBoost and how it influences model convergence.

8.4 Ans: The learning rate in boosting algorithms, such as AdaBoost and XGBoost, controls the impact of each subsequently added weak learner on the final model. A smaller learning rate means that each individual learner contributes more subtly to the final model, requiring more learners to converge but typically leading to a more robust and generalized model. Conversely, a larger learning rate allows the model to converge faster, with fewer learners, but increases the risk of overfitting. Proper tuning of the learning rate is crucial for balancing the trade-off between convergence speed and model accuracy.

8.5 Explain the role of “Loss Functions” in the formulation of boosting algorithms, specifically referring to Gradient Boosting and XGBoost.

8.5 Ans: In boosting algorithms like Gradient Boosting and XGBoost, the choice of loss function is critical for directing the training process. These algorithms minimize a specified loss function iteratively by adding weak learners that reduce the residual errors. For instance, Gradient Boosting can use loss functions like squared error for regression or logistic loss for classification, which directly influences the algorithm's approach to reducing prediction errors. XGBoost extends this by allowing custom loss functions, thus enabling more tailored model training that is optimized for specific types of data and prediction tasks.

8.6 Write down the mathematical expression for the general aggregation strategy used in ensemble learning as discussed in the chapter. Explain how this strategy works in the context of both Bagging and Boosting.

8.6 Ans: **Formulation and Explanation:** In ensemble methods, the general model output can be represented as

$$F(\mathbf{x}) = \Phi \left( \sum_{m=1}^M \alpha_m f_m(\mathbf{x}) \right),$$

where  $\Phi$  is a function translating aggregated model outputs into the final prediction,  $\alpha_m$  are the weights of individual models, and  $f_m(\mathbf{x})$  are the predictions from individual models.

- **In Bagging**, each model has an equal weight,  $\alpha_m = \frac{1}{M}$ , reflecting the democratic voting or averaging principle among models, which effectively reduces variance.
- **In Boosting**, weights  $\alpha_m$  are adjusted according to the model's performance, emphasizing models that correct the mistakes of predecessors, thus reducing both bias and variance in a sequential manner.

8.7 Provide a derivation of how bagging helps reduce variance without significantly affecting bias (assuming independence of individual models). Refer to the mathematical formulation of a bagged estimator and discuss its implications on a model's prediction error.

8.7 Ans: **Derivation and Explanation:** Consider a model's prediction as  $f(\mathbf{x})$  and suppose the model's bias and variance are given by

$$\text{Bias}(f(\mathbf{x})) = \mathbb{E}[f(\mathbf{x})] - f_{\text{true}}(\mathbf{x}),$$

$$\text{Variance}(f(\mathbf{x})) = \mathbb{E}[(f(\mathbf{x}) - \mathbb{E}[f(\mathbf{x})])^2].$$

In bagging, the aggregated model prediction is

$$F(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M f_m(\mathbf{x}),$$

where each  $f_m(\mathbf{x})$  is trained on a different bootstrap sample. Assuming the models are independent,

$$\text{Bias}(F(\mathbf{x})) = \text{Bias}(f(\mathbf{x})),$$

$$\text{Variance}(F(\mathbf{x})) = \frac{1}{M} \text{Variance}(f(\mathbf{x})).$$

The derivation shows that while the bias remains the same as that of any individual model, the variance is reduced by a factor of  $M$ , the number of models in the ensemble, demonstrating how bagging effectively decreases variance.

- 8.8 Derive the weight update rule used in the AdaBoost algorithm as mentioned in the chapter. Explain how this formula helps to focus learning on more difficult training examples.

8.8 Ans: **Derivation and Explanation:** AdaBoost aims to minimize an exponential loss function by updating the weights of training examples at each iteration. At iteration  $m$ , AdaBoost fits a weak classifier  $h_m$  to the weighted data, where each weight  $w_{i,m}$  is associated with the  $i$ -th training example. The error rate  $\epsilon_m$  of  $h_m$  is calculated as follows:

$$\epsilon_m = \frac{\sum_{i=1}^N w_{i,m} \mathbf{1}(y_i \neq h_m(x_i))}{\sum_{i=1}^N w_{i,m}},$$

where  $\mathbf{1}(\cdot)$  is the indicator function, activated when  $y_i \neq h_m(x_i)$ , and  $N$  is the total number of training examples.

Given  $\epsilon_m$ , the weight update coefficient  $\alpha_m$  for the classifier  $h_m$  is derived by aiming to balance the weights of correctly and incorrectly classified examples after the update:

$$w_{i,m+1} = w_{i,m} e^{-\alpha_m y_i h_m(x_i)},$$

where  $y_i$  is the true label, and  $h_m(x_i)$  is the prediction. This formulation ensures that weights are increased for incorrectly classified instances ( $y_i h_m(x_i) = -1$ ) and decreased for correctly classified instances ( $y_i h_m(x_i) = 1$ ).

To find  $\alpha_m$ , consider the goal of keeping the sum of weights constant before and after updating:

$$\sum_{i: y_i = h_m(x_i)} w_{i,m} e^{-\alpha_m} + \sum_{i: y_i \neq h_m(x_i)} w_{i,m} e^{\alpha_m} = \sum_{i=1}^N w_{i,m}.$$

Substituting and rearranging gives:

$$(1 - \epsilon_m)e^{-\alpha_m} + \epsilon_me^{\alpha_m} = 1.$$

Solving for  $\alpha_m$  yields:

$$\alpha_m = \frac{1}{2} \ln \left( \frac{1 - \epsilon_m}{\epsilon_m} \right).$$

This reweighting mechanism incentivizes the AdaBoost algorithm to concentrate subsequent training on the examples that are more challenging to classify, progressively refining the model's accuracy on these difficult cases.

8.9 Demonstrate mathematically how boosting can achieve lower bias with an increasing number of weak learners.

8.9 Ans: The ensemble model in boosting after  $M$  iterations is

$$F(\mathbf{x}) = \sum_{m=1}^M \alpha_m h_m(\mathbf{x}),$$

Assuming each weak learner corrects the residual errors from the previous model, the expected prediction of the ensemble becomes an increasingly accurate approximation of the true function  $f(\mathbf{x})$ , leading to

$$\text{Bias}(F(\mathbf{x})) = |\mathbb{E}[F(\mathbf{x})] - f(\mathbf{x})|.$$

As  $M$  increases, the collective corrections of weak learners reduce this bias, enhancing the ensemble's accuracy.

8.10 Implement a basic Random Forest classifier using Python's scikit-learn library on the Iris data set. Demonstrate the classifier's training process, prediction, and evaluate its accuracy on the test set.

8.10 Ans: **Solution:**

```
from sklearn.datasets import load_iris
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# Load the Iris dataset
data = load_iris()
X, y = data.data, data.target

# Split the data into training and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y,
                                                    test_size=0.3, random_state=42)

# Initialize and train the Random Forest classifier
```

```

clf = RandomForestClassifier(n_estimators=100)
clf.fit(X_train, y_train)

# Make predictions on the test set
predictions = clf.predict(X_test)

# Evaluate the accuracy of the classifier
accuracy = accuracy_score(y_test, predictions)
print("Accuracy:", accuracy)

```

## Chapter 9

### Performance Evaluation and Statistical Inference

#### Exercises

9.1 Suppose you are given a data set of 800 samples in total. Which of the following is/are among the best for the given options for evaluating an algorithm based on this data set?

- (a) Use all samples as training and test sets.
- (b) Based on single hold-out test, i.e., use  $M$  samples for testing and use  $800 - M$  samples for training the algorithm.
- (c) Based on multiple hold-out tests, i.e., perform hold-out test multiple times using randomly partitioned training and test sets.
- (d) Based on  $N$ -fold cross-validation test.
- (e) Based on leave-one-out test, i.e., use 1 sample for testing and use 799 samples for training the algorithm, and repeat this procedure until all samples have been tested. The final results are averaged.

9.1 Ans: (c), (d) and (e)

- (a) This option does not consider untrained samples for evaluation.
- (b) This option is subjected to biased partitioning.
- (c) This option can reduce the partitioning bias.
- (d) This option can also minimize partitioning bias and it uses all data for testing.
- (e) This option is among the most thorough ones but it consumes heavy training effort for large data sets.

9.2 How can the hold-out test be performed using Python?

9.2 Ans: The hold-out test can be implemented based on the `train_test_split` function from the `sklearn.model_selection` module. This function takes in data as a NumPy array, and it returns two new arrays: one for the training data and one for the testing data.

9.3 What is the difference between regression error and classification error?

Provide examples of their score measurements.

- 9.3 Ans: • Regression error refers to the aggregation of distance difference between each sample output and the target output where the distance can be measured using squared error distance or absolute error distance. Suppose  $y_i$  denotes the sample target and  $\hat{y}_i$  denotes the sample output (estimation) and  $m$  is the number of samples, then the mean squared error (MSE) can be written as  $\frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)^2$ , and the mean absolute error can be written as  $\frac{1}{m} \sum_{i=1}^m |\hat{y}_i - y_i|$ .
- Classification error refers to the aggregation of samples classified incorrectly which can be expressed as a classification error count or as an error ratio. The classification error ratio in general is the total number of misclassified samples over total number of samples, i.e.,  $(TP+TN)/(n^+ + n^-)$  (See Table 4.3).

9.4 How to do a hyperparameter tuning based on grid search in Python?

- 9.4 Ans: Grid search involves a systematic testing of different combinations of hyperparameter values in order to find the combination with best validation performance for the model. In Python, the GridSearchCV module from scikit-learn library can be utilized to perform the grid search under the cross-validation mode.

9.5 Under what situation cross-validation should be avoided?

- 9.5 Ans: • When the data set is too small for splitting to get enough training size.
- When data is not independent and identically distributed.
- For time series data when data leakage occurs. In other words, data not intended for training could leak out and be used to train the model.

9.6 Is it possible to check if a classifier has underfit or overfit the data using only the training set? If yes, how? If no why?

- 9.6 Ans: Yes, it is possible to use only training set to check whether underfit or overfit occurs in the classifier model. The training set can be split into training and validation sets and use the validation set for checking. If the training accuracy is much higher than the validation accuracy, then it is likely that the model has overfit the training data. If the training accuracy is much lower than the validation accuracy, then it is likely that the model has underfit the training data.

9.7 A classifier that attains 100% accuracy on the training set and 70% accuracy on test set is better than another classifier that attains 80% accuracy on the training set and 75% accuracy on test set. True or False?

- 9.7 Ans: False (Reason: The 100% training accuracy with 70% test accuracy appears to have overfit the data.)

9.8 Suppose the hyperparameter  $\eta$  has been selected for a model using tenfold cross-validation. Among the following, the best way to decide a final model to use and estimate its error is to

- (a) pick any of the 10 models from the 10 folds as the final model; use the average CV error for the 10 models as its error estimate.

- (b) pick any of the 10 models from the 10 folds as the final model; use its error estimate on the held-out data.
- (c) train a new model on the full data set using the hyperparameter  $\eta$  obtained; use the average CV error as its error estimate.
- (d) average all of the 10 models from the 10 folds; use the average CV error as its error estimate.
- (e) average all of the 10 models from the 10 folds; use the error of the combined model on the full training set.

9.8 Ans: (c)

1. It is best to utilize all the training data to find the final model.
2. As there is no mention of validation set or test set availability, the only best error estimate is the average CV error.

9.9 The \_\_\_\_\_ the  $p$ -value, the stronger the evidence against the null hypothesis provided by the data.

- (a) larger
- (b) smaller

9.9 Ans: (b)

According to <https://en.wikipedia.org/wiki/P-value>: “In null hypothesis significance testing, the  $p$ -value is the probability of obtaining test results at least as extreme as the result actually observed, under the assumption that the null hypothesis is correct. A very small  $p$ -value means that such an extreme observed outcome would be very unlikely under the null hypothesis”.

9.10 One-sample  $t$ -test: To test the hypothesis that preparation study for more than 3 h for three days right before the test helps to score over 70 marks. Here are the results of a random sample of 12 students who had done the preparation study before the test: 86, 81, 71, 90, 69, 64, 76, 85, 77, 71, 80, 62. Test using the following hypotheses, report the test statistic with the  $p$ -value at 95% confidence level, then summarize your conclusion.

9.10 Ans: Hypothesis:

$H_0$ : marks  $\mu = 70$  (no effect: preparation study for 3 d does not help increase the mean mark)

$H_1$ : marks  $\mu \neq 70$  ( $> 70$ ) (effect: preparation study for 3 d helps increase the mean mark)

Test statistic:

From the data, we obtain the mean ( $\bar{x}$ ) = 76 and

$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} = 8.8318$ . Then we get

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{76 - 70}{8.8318/\sqrt{12}} = \frac{6}{2.5495} = 2.3534$$

